QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/152B
21 JULY 1986
SUPERSEDING
MIL-M-38510/152A
18 March 1985

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,
DATA DECODERS/DEMULTIPLEXERS,
MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, data decoders/demultiplexers logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit
01	4-line to 16-line decoder/demultiplexer
02	Dual 2-line to 4-line decoder/demultiplexer
08	Dual 2-line to 4-line decoder/demultiplexer, open collector
04	Binary-to-octal decoder
05	BCD-to-decimal decoder
06	BCD-to-decimal decoder

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
J	D-3 (24-lead, 1/2" x 1 1/4"), dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8"), flat package
Z	F-8 (24-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to 5.5 V dc
Storage temperature range	-65°C to +150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Maximum power dissipation per gate (P_D) 1/:	
Device type 01	270 mW
Device type 02	195 mW
Device type 03	195 mW
Device type 04	140 mW
Device type 05	190 mW
Device type 06	190 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, D, E, F, J, K and Z	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 3/:	
At low logic level	10 maximum
At high logic level 4/	20 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ Maximum junction temperature shall not be exceeded except for short duration burn-in screening conditions per method 5004 of MIL-STD-883.

3/ Device will fanout in both high and low levels to the specified number of inputs on the same device type as that being tested.

4/ A fanout of 20 normalized loads is provided to facilitate connection of unused inputs to used inputs.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -0.8 mA	01, 02, 04, 05, 06	2.4		V
High level output leakage current	I_{CEX}	V_{CC} = 4.5 V, V_{OUT} = 5.5 V	02		250	µA
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA	All		-1.5	V
Low level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01, 02, 03, 06	-0.7	-1.6	mA
			04, 05	-0.4	-1.3	mA
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	All		60	µA
High level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		150	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = 0 V 1/	02	-20	-55	mA
			04,05,06	-10	-55	mA
			01	-20	-70	mA

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Supply current	I_{CC}	V_{CC} = 5.5 V	01		62	mA
			02, 03		35	mA
			04		25	mA
			05		34	mA
			06		44	mA
Propagation delay time high-to-low level output from A, B, C, or D to output	t_{PHL1}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 5)	01	5	49	ns
Propagation delay time low-to-high level output from A, B, C, or D to output	t_{PLH1}		01	5	53	ns
Propagation delay time high-to-low level output from G1 or G2 to output	t_{PHL2}		01	5	42	ns
Propagation delay time low-to-high level output from G1 or G2 to output	t_{PLH2}		01	5	45	ns
Propagation delay time low-to-high level output from 1G or 2G to output	t_{PLH1}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 6)	02	3	33	ns
Propagation delay time high-to-low level output from 1G or 2G to output	t_{PHL1}		02	3	42	ns
Propagation delay time low-to-high level output A or B to output through two levels	t_{PLH2}		02	3	33	ns
Propagation delay time high-to-low level output A or B to output through two levels	t_{PHL2}		02	3	42	ns
Propagation delay time low-to-high level output 2C to output	t_{PLH3}		02	3	33	ns
Propagation delay time high-to-low level output 2C to output	t_{PHL3}		02	3	42	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay time low-to-high level output 1C to output	t_{PLH4}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 6)	02	3	38	ns
Propagation delay time high-to-low level output 1C to output	t_{PHL4}		02	3	46	ns
Propagation delay time low-to-high level output A or B to output	t_{PLH5}		02	3	48	ns
Propagation delay time high-to-low level output A or B to output	t_{PHL5}		02	3	48	ns
Propagation delay time low-to-high level output from 1G or 2G to output	t_{PLH1}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 7)	03	3	43	ns
Propagation delay time high-to-low level output from 1G or 2G to output	t_{PHL1}		03	3	52	ns
Propagation delay time low-to-high level output A or B to output through two levels	t_{PLH2}		03	3	43	ns
Propagation delay time high-to-low level output A or B to output through two levels	t_{PHL2}		03	3	52	ns
Propagation delay time low-to-high level output 2C to output	t_{PLH3}		03	3	43	ns
Propagation delay time high-to-low level output 2C to output	t_{PHL3}		03	3	52	ns
Propagation delay time low-to-high level output 1C to output	t_{PLH4}		03	3	48	ns
Propagation delay time high-to-low level output 1C to output	t_{PHL4}		03	3	56	ns
Propagation delay time low-to-high level output A or B to output	t_{PLH5}		03	3	57	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time high-to-low level output A or B to output	t_{PHL5}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 7)	03	3	57	ns
Propagation delay time high-to-low level output A, B, C, or D to output through two levels	t_{PHL1}	R_L = 390Ω ±5% C_L = 50 pF minimum ±10% (figure 8)	04, 05, 06	5	48	ns
Propagation delay time low-to-high level output A, B, C, or D to output through two levels	t_{PLH1}		04, 05, 06	5	48	ns
Propagation delay time high-to-low level output A, B, C, or D to output	t_{PHL2}		04, 05, 06	5	53	ns
Propagation delay time low-to-high level output A, B, C, or D to output	t_{PLH2}		04, 05, 06	5	53	ns

1/ Not more than one output should be shorted at one time.

TABLE II. Electrical test requirements.

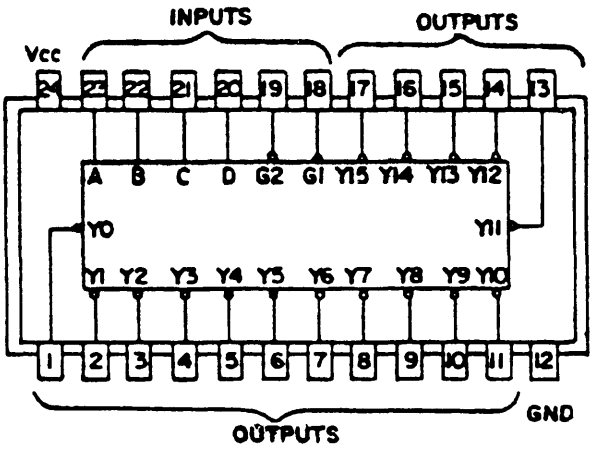
MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	---
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

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Device type 01

Cases J, K, and Z



Device types 02 and 03

Cases E and F

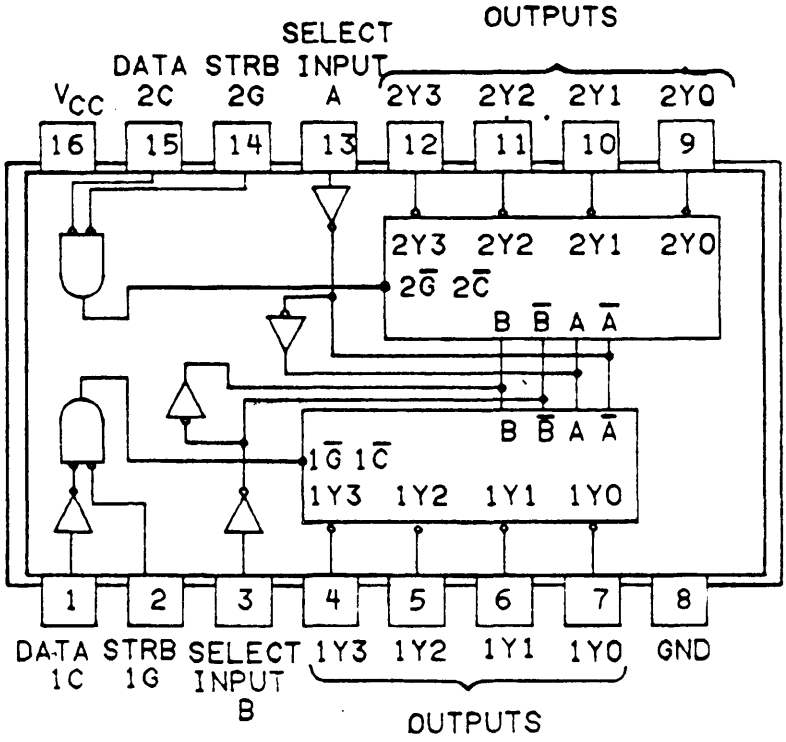
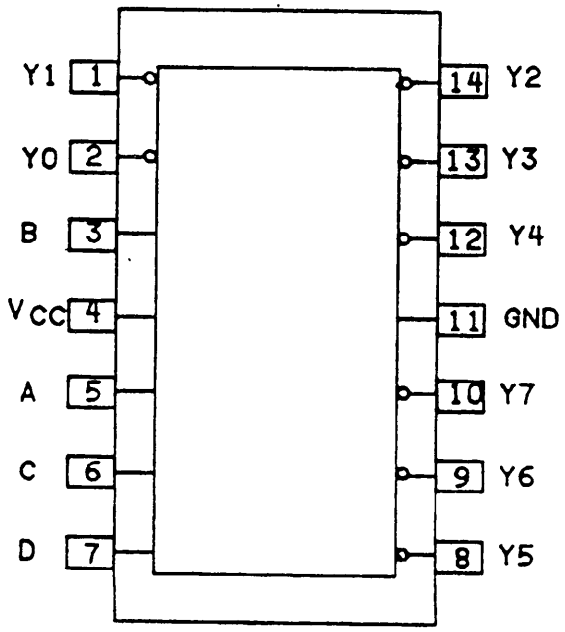


FIGURE 1. Terminal connections (top view).

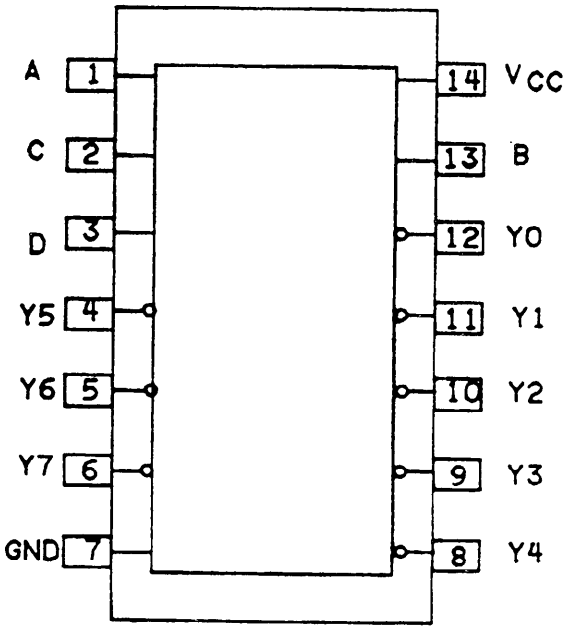
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Device type 04

Cases A, B, and D



Case C



Device types 05 and 06

Cases E and F

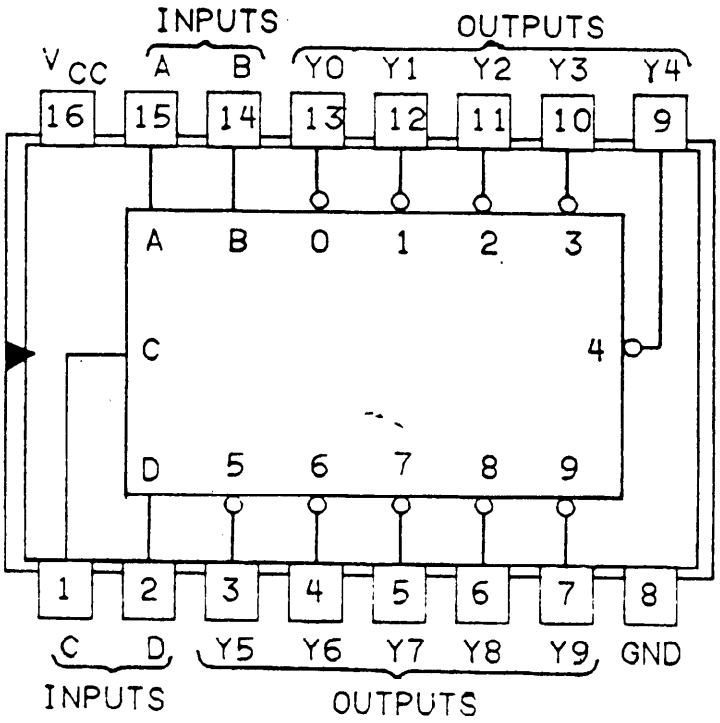


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

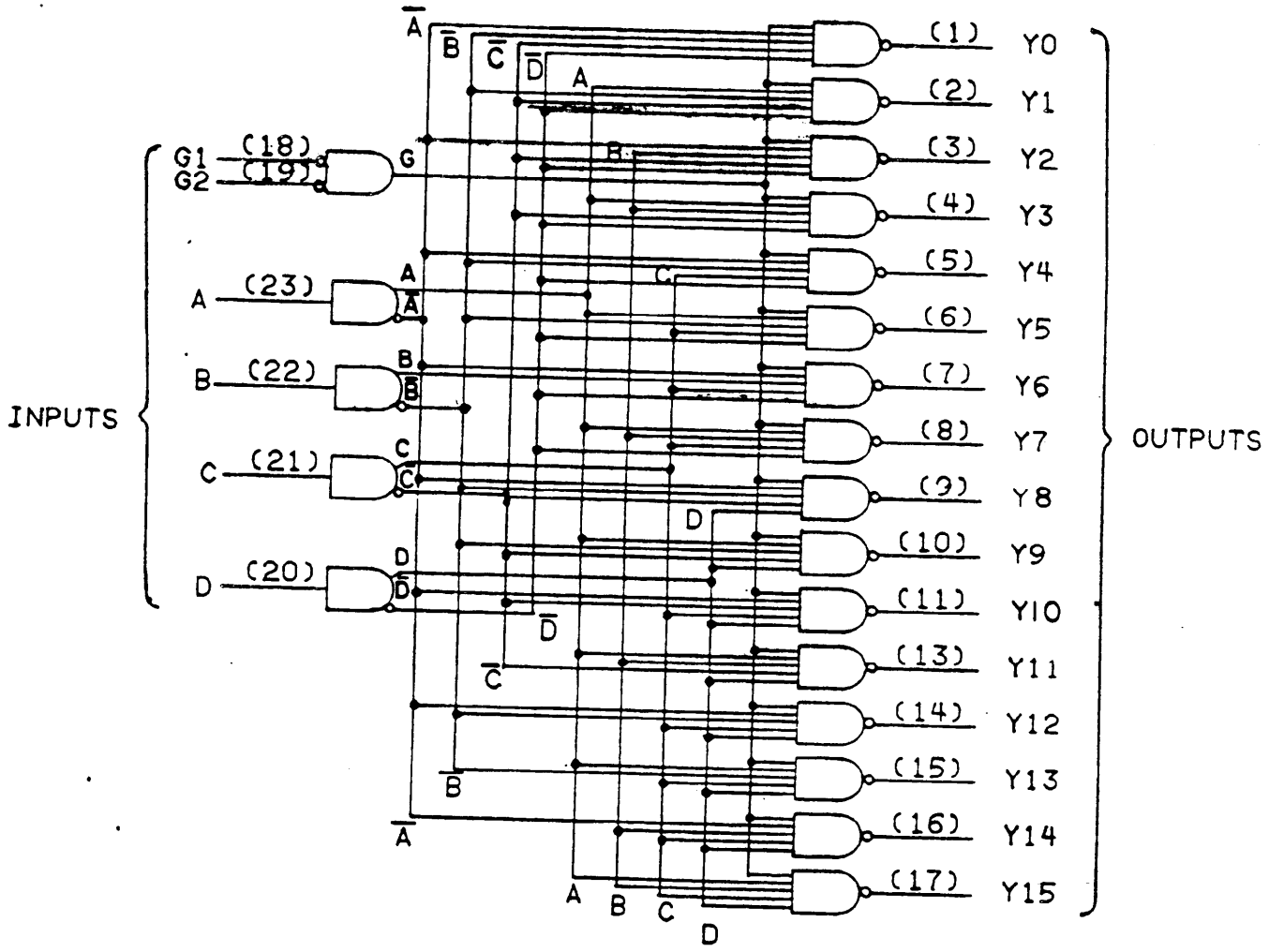


FIGURE 2. Logic diagrams.

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Device types 02 and 03

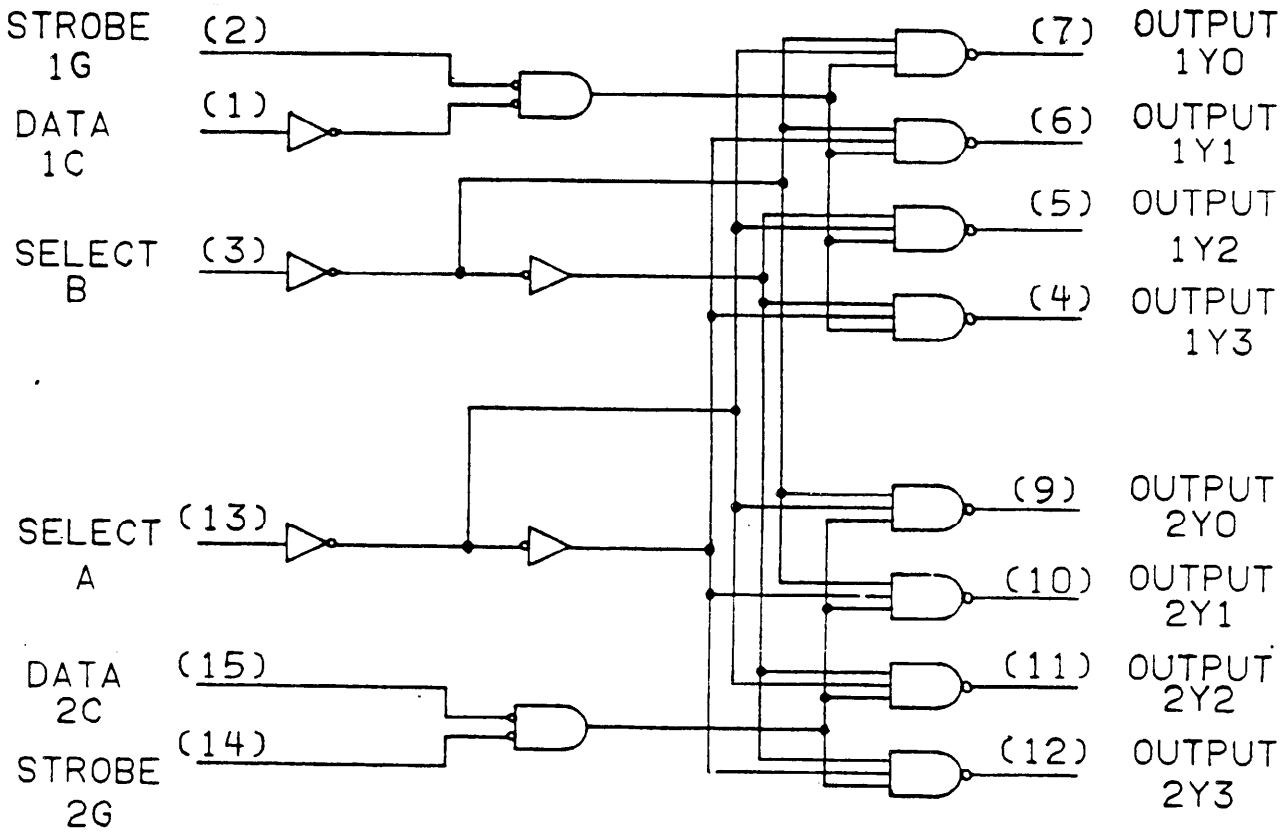
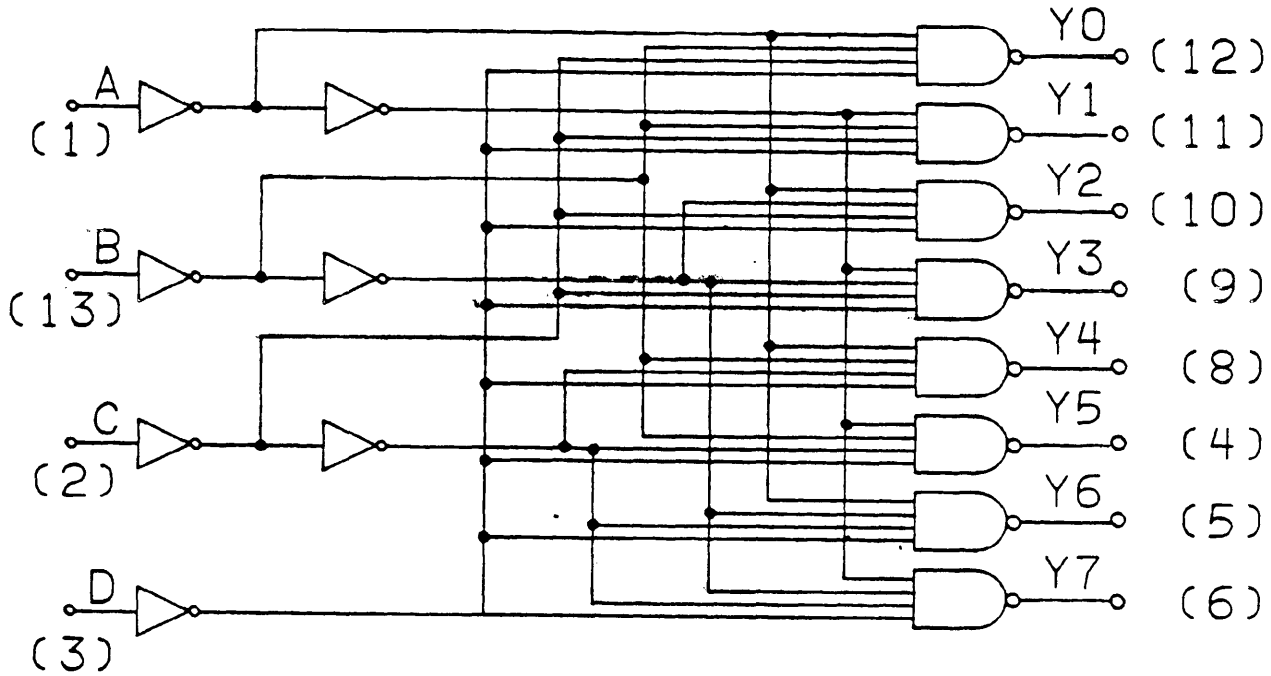


FIGURE 2. Logic diagrams - Continued.

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Device type 04



Device type 05

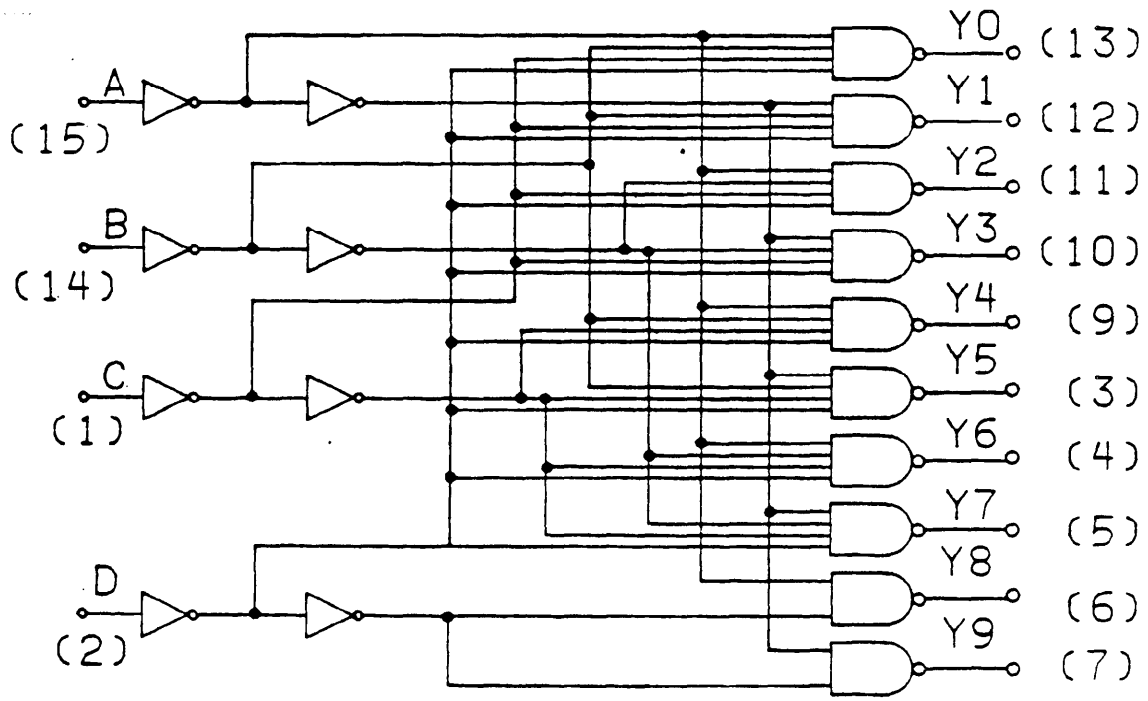


FIGURE 2. Logic diagrams - Continued.

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Device type 06

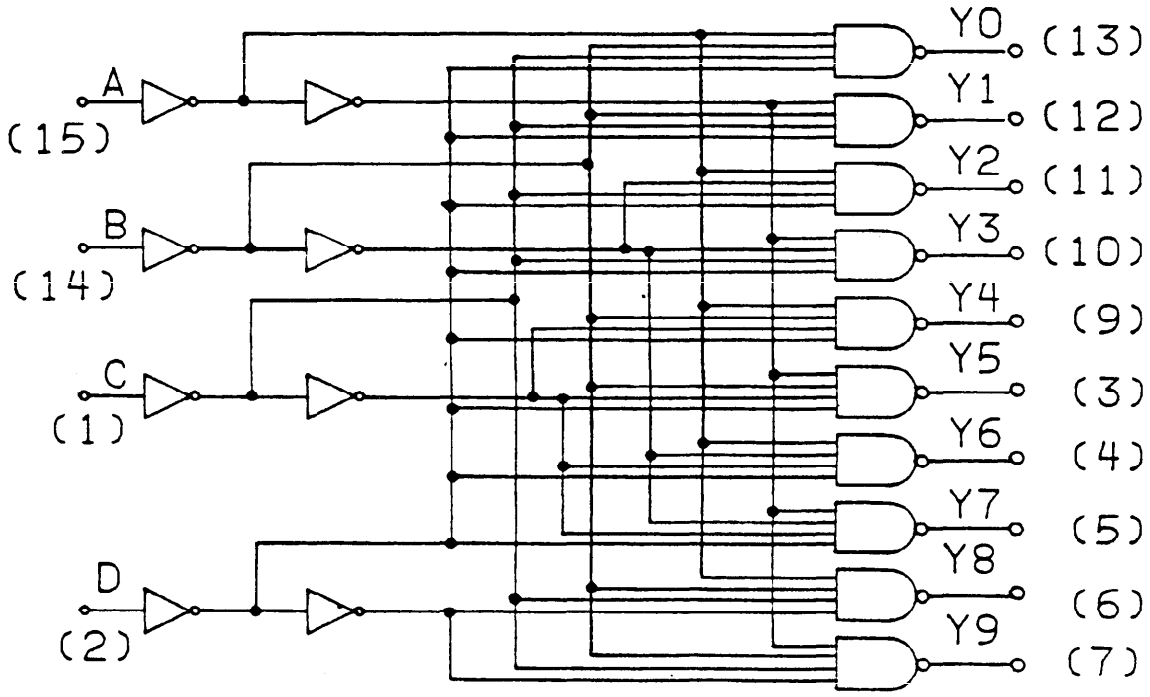


FIGURE 2. Logic diagrams - Continued.

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Device type 01

Inputs						Outputs															
G1	G2	D	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L
L	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	L	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High, L = Low, X = Irrelevant.

Device types 02 and 03

2-LINE TO 4-LINE DECODER OR 1-LINE TO 4-LINE DEMULTIPLEXER

Inputs				Outputs				Inputs				Outputs			
Select		Strobe	Data					Select		Strobe	Data				
B	A	1G	1C	1Y0	1Y1	1Y2	1Y3	B	A	2G	2C	2Y0	2Y1	2Y2	2Y3
X	X	H	X	H	H	H	H	X	X	H	X	H	H	H	H
L	L	L	H	L	H	H	H	L	L	L	L	L	H	H	H
L	H	L	H	H	L	H	H	L	H	L	L	H	L	H	H
H	L	L	H	H	H	L	H	H	L	L	L	H	H	L	H
H	H	L	H	H	H	H	L	H	H	L	L	H	H	H	L
X	X	X	L	H	H	H	H	X	X	X	H	H	H	H	H

FIGURE 3. Truth tables.

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Device types 02 and 03

3-line to 8-line decoder or 1-line to 8-line demultiplexer

Inputs				Outputs							
Select			Strobe or Data	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
C†	B	A	G†	2Y0	2Y1	2Y2	2Y3	1Y0	1Y1	1Y2	1Y3
X	X	X	H	H	H	H	H	H	H	H	H
L	L	L	L	L	H	H	H	H	H	H	H
L	L	H	L	H	L	H	H	H	H	H	H
L	H	L	L	H	H	L	H	H	H	H	H
L	H	H	L	H	H	H	L	H	H	H	H
H	L	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	L	H	H
H	H	L	L	H	H	H	H	H	H	L	H
H	H	H	L	H	H	H	H	H	H	H	L

†C = Inputs 1C and 2C connected together
†G = Inputs 1G and 2G connected together
H = High level, L = Low level, X = Irrelevant

Device type 04

Inputs				Outputs							
A	B	C	D	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	L	H	H	H	H	H	H
L	H	L	L	H	H	L	H	H	H	H	H
H	H	L	L	H	H	H	L	H	H	H	H
L	L	H	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	L	H	H
L	H	H	L	H	H	H	H	H	H	L	H
H	H	H	L	H	H	H	H	H	H	H	L
L	L	L	H	H	H	H	H	H	H	H	H
H	L	L	H	H	H	H	H	H	H	H	H
L	H	L	H	H	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H	H	H	H
L	L	H	H	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H
H	H	H	H	H	H	H	H	H	H	H	H

H = High voltage level
L = Low voltage level

FIGURE 3. Truth tables - Continued.

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Device type 05

Inputs				Outputs									
A	B	C	D	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9
L	L	L	L	L	H	H	H	H	H	H	H	H	H
H	L	L	L	H	L	H	H	H	H	H	H	H	H
L	H	L	L	H	H	L	H	H	H	H	H	H	H
H	H	L	L	H	H	H	L	H	H	H	H	H	H
L	L	H	L	H	H	H	H	L	H	H	H	H	H
H	L	H	L	H	H	H	H	H	L	H	H	H	H
L	H	H	L	H	H	H	H	H	H	L	H	H	H
H	H	H	L	H	H	H	H	H	H	H	L	H	H
L	L	L	H	H	H	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L
L	H	L	H	H	H	H	H	H	H	H	H	L	H
H	H	L	H	H	H	H	H	H	H	H	H	H	L
L	L	H	H	H	H	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	H	L
L	H	H	H	H	H	H	H	H	H	H	H	L	H
H	H	H	H	H	H	H	H	H	H	H	H	H	L

H = High voltage level
L = Low voltage level

Device type 06

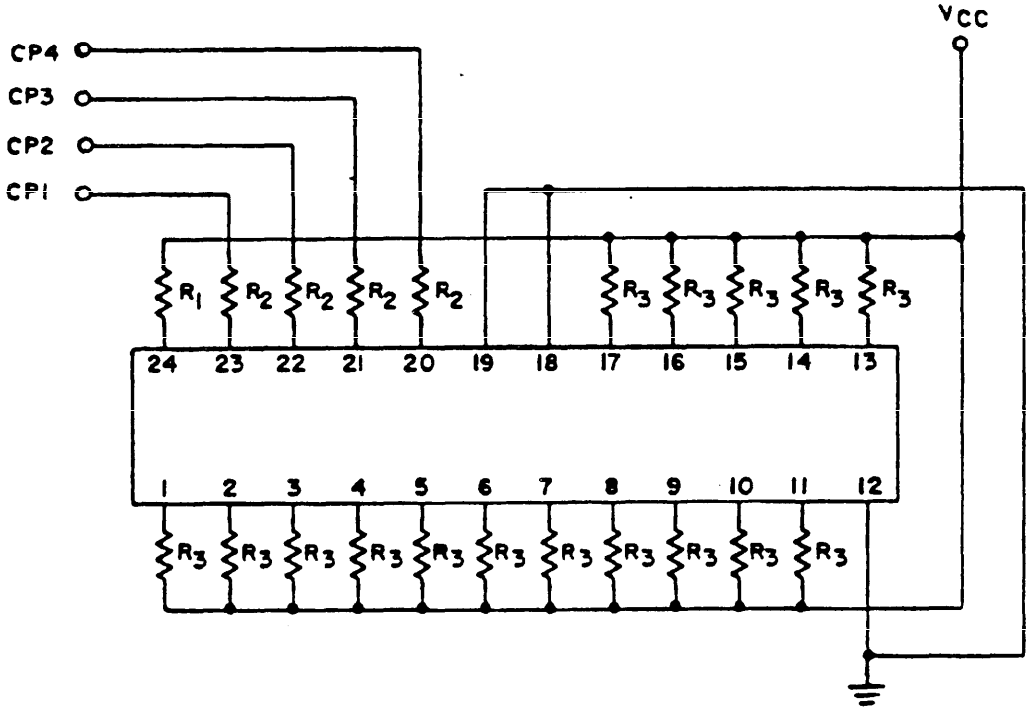
Inputs				Outputs									
A	B	C	D	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9
L	L	L	L	L	H	H	H	H	H	H	H	H	H
H	L	L	L	H	L	H	H	H	H	H	H	H	H
L	H	L	L	H	H	L	H	H	H	H	H	H	H
H	H	L	L	H	H	H	L	H	H	H	H	H	H
L	L	H	L	H	H	H	H	L	H	H	H	H	H
H	L	H	L	H	H	H	H	H	L	H	H	H	H
L	H	H	L	H	H	H	H	H	H	L	H	H	H
H	H	H	L	H	H	H	H	H	H	H	L	H	H
L	L	L	H	H	H	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L
L	H	L	H	H	H	H	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H	H	H	H	H	H
L	L	H	H	H	H	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H	H	H
H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High voltage level
L = Low voltage level

FIGURE 3. Truth tables - Continued.

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Device type 01



Device types 02 and 03

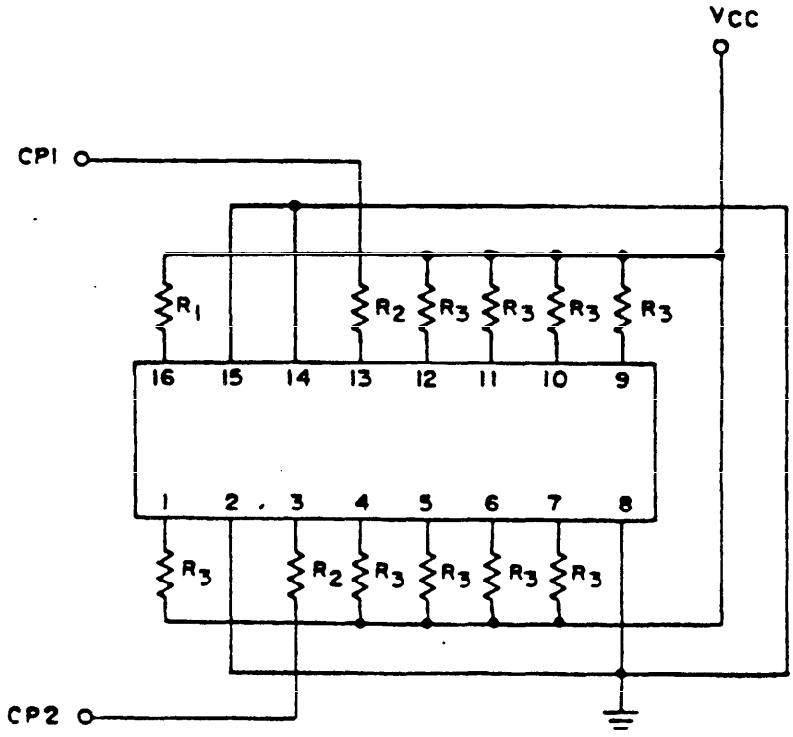


FIGURE 4. Burn-in and life test circuit.

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Device type 04

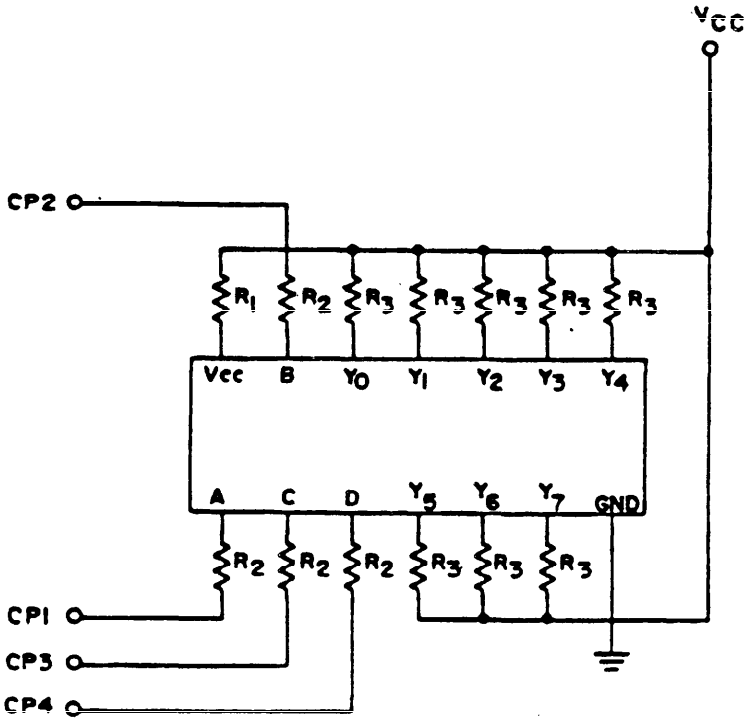
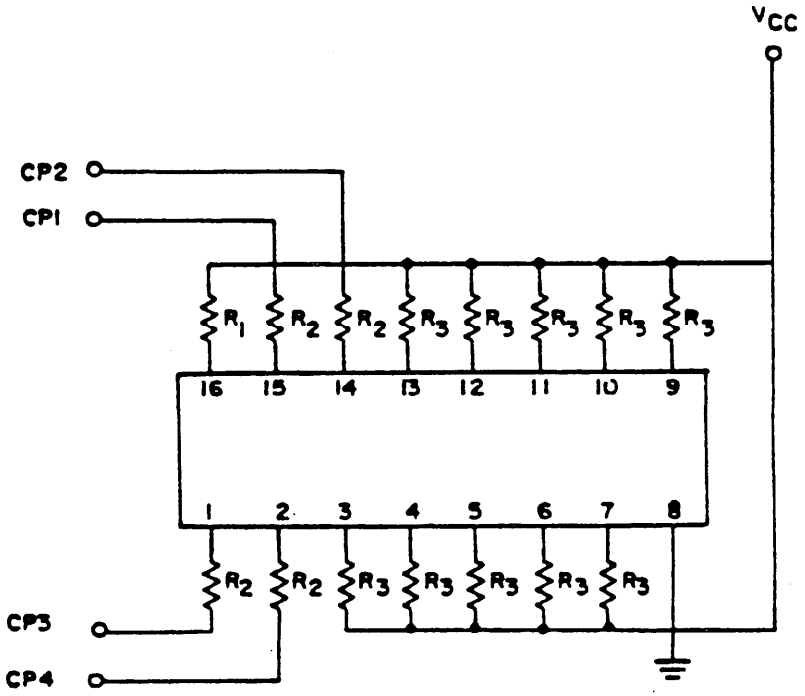


FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 05 and 06



NOTES:

1. V_{CC} is 5.0 V minimum at the device terminal.
2. R_1 = 2.0Ω ±5%; R_2 = 27Ω ±5%; R_3 = 270Ω ±5%.
3. CP1 = 100 kHz ±25% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.

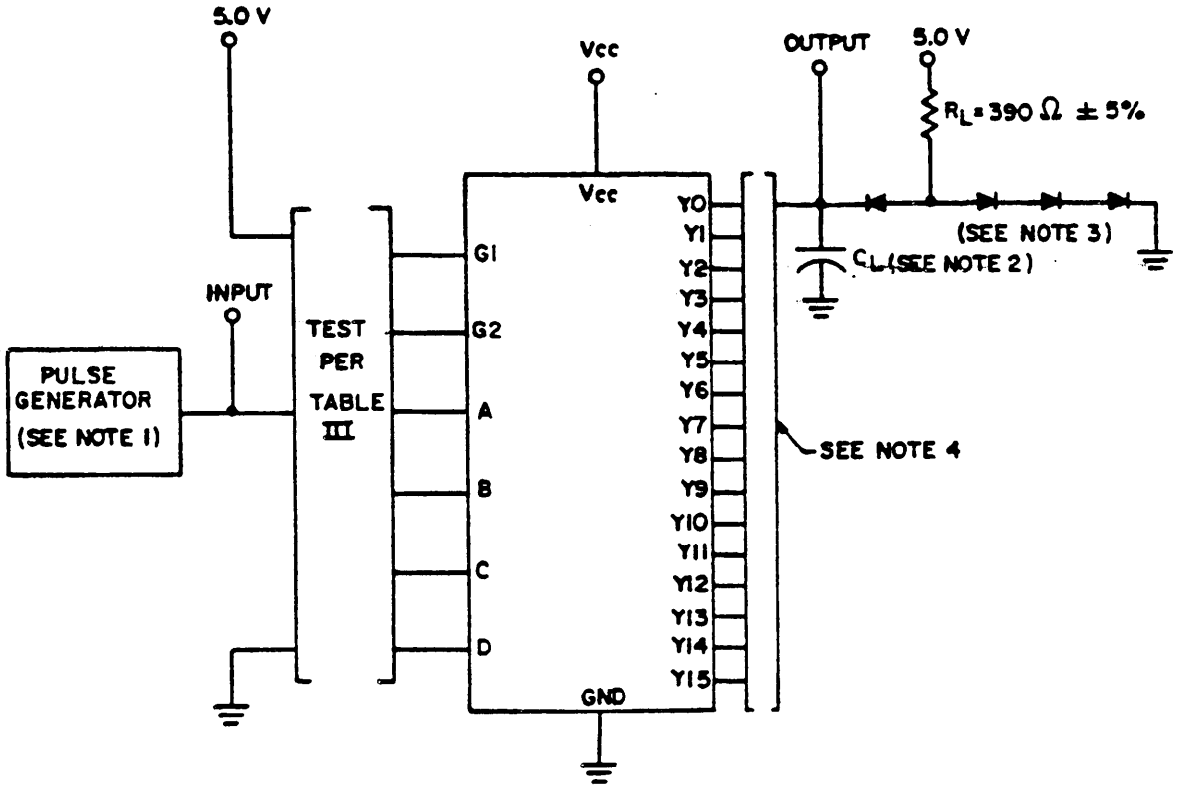
   CP2 = 50 kHz ±25% square wave synchronized with CP1; all other conditions same as for CP1.

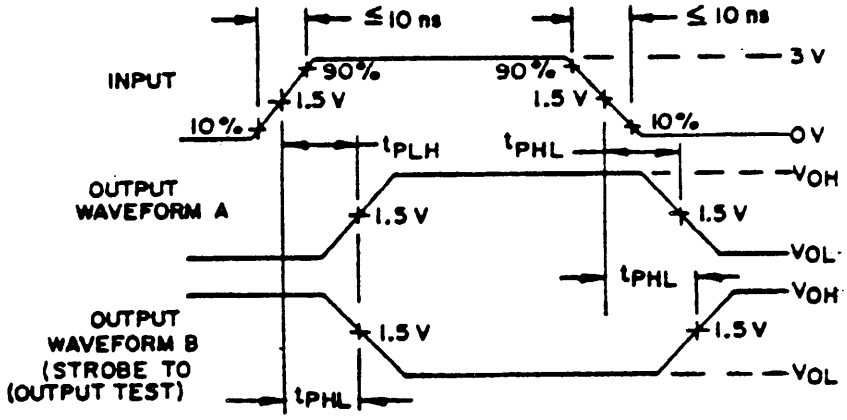
   CP3 = 25 kHz ±25% square wave synchronized with CP2; all other conditions same as CP1.

   CP4 = 12.5 kHz ±25% square wave synchronized with CP3; all other conditions same as CP1.

FIGURE 4. Burn-in and life test circuit - Continued.

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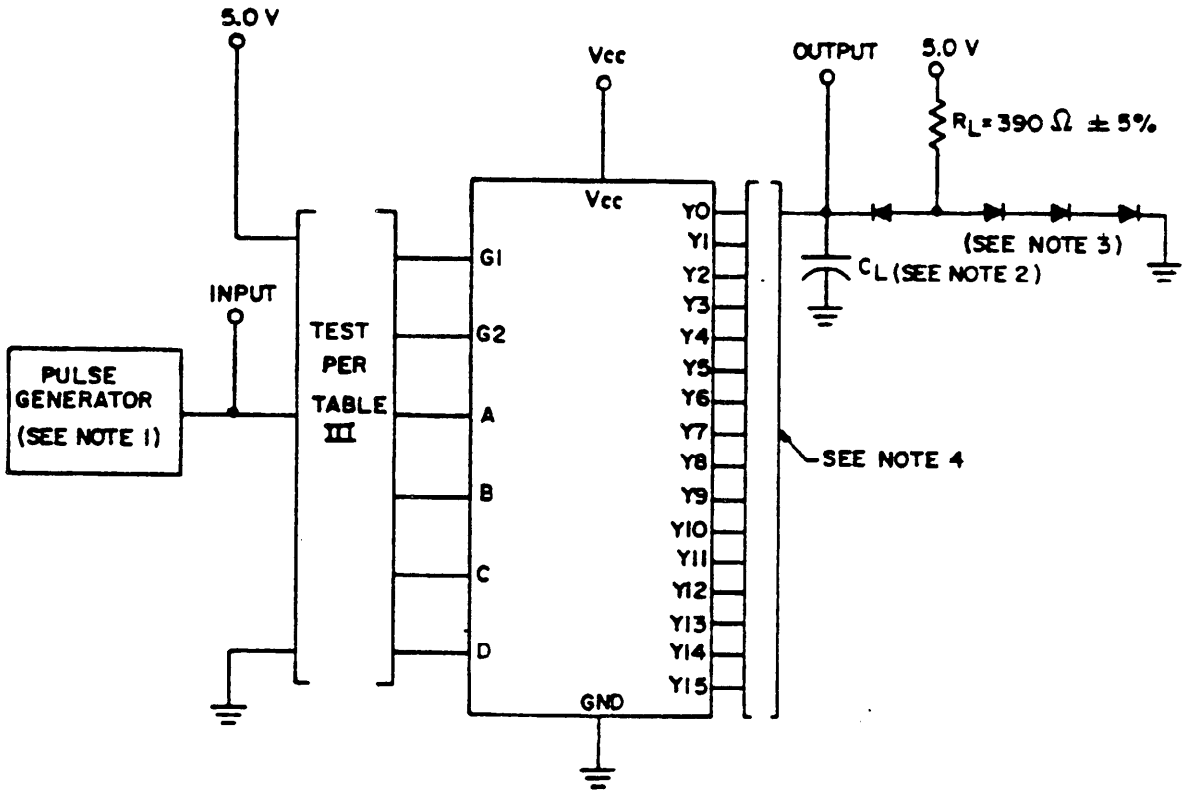


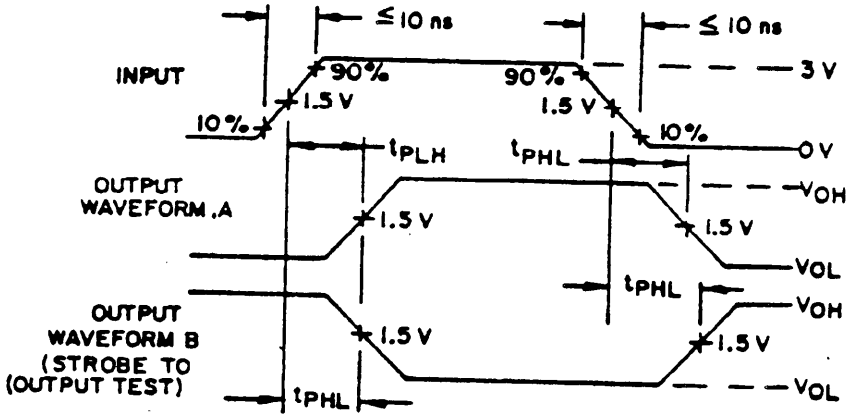


NOTES:
1. The pulse generator has the following characteristics: PRR ≤ 1 MHz, duty cycle = 50%, and Z_{out} ≈ 50Ω.
2. C_L = 50 pF ±10% minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 5. Switching test for device type 01.

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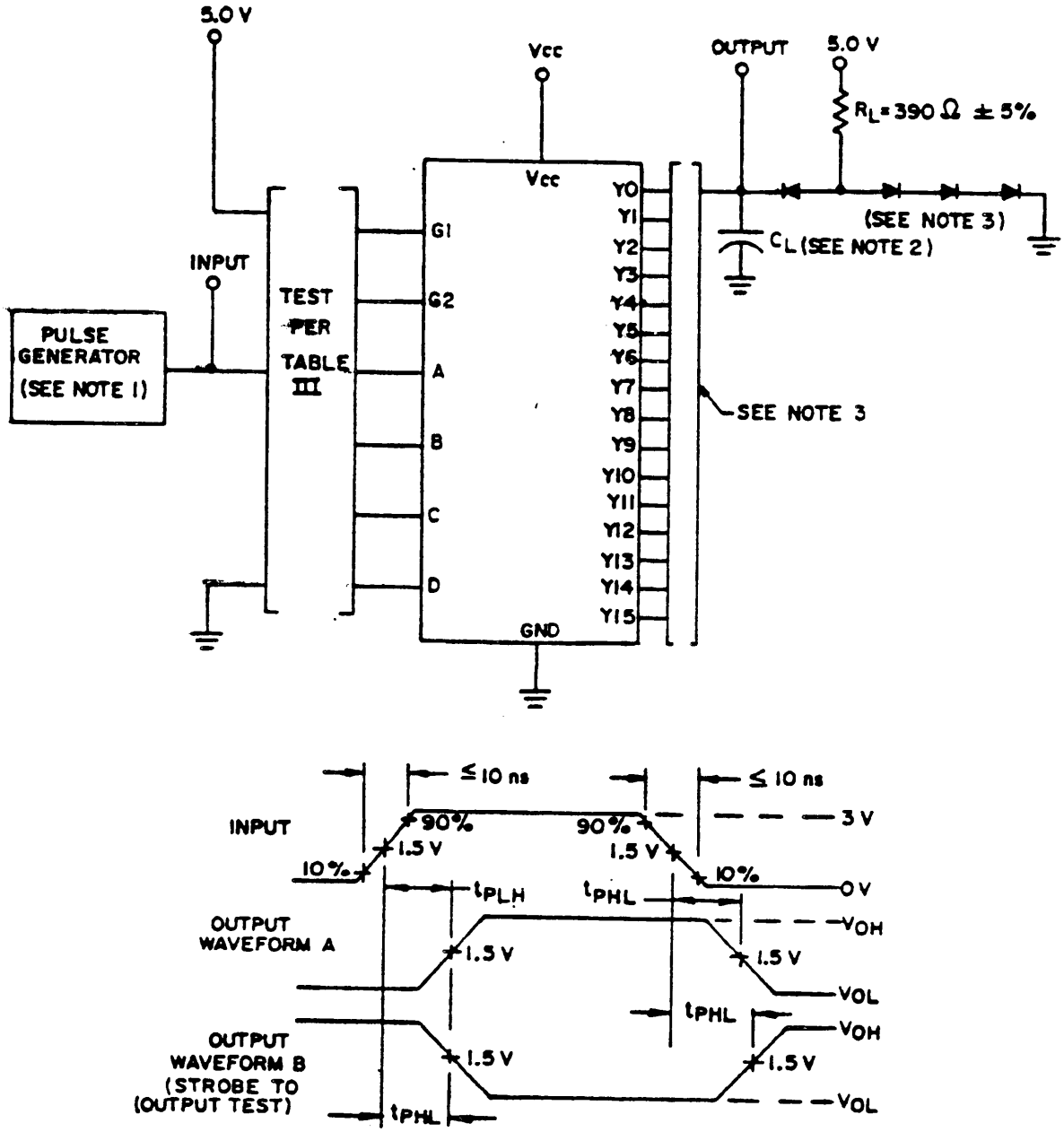




NOTES:
1. The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50%, and Z_{out}=50Ω.
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 6. Switching test for device type 02.

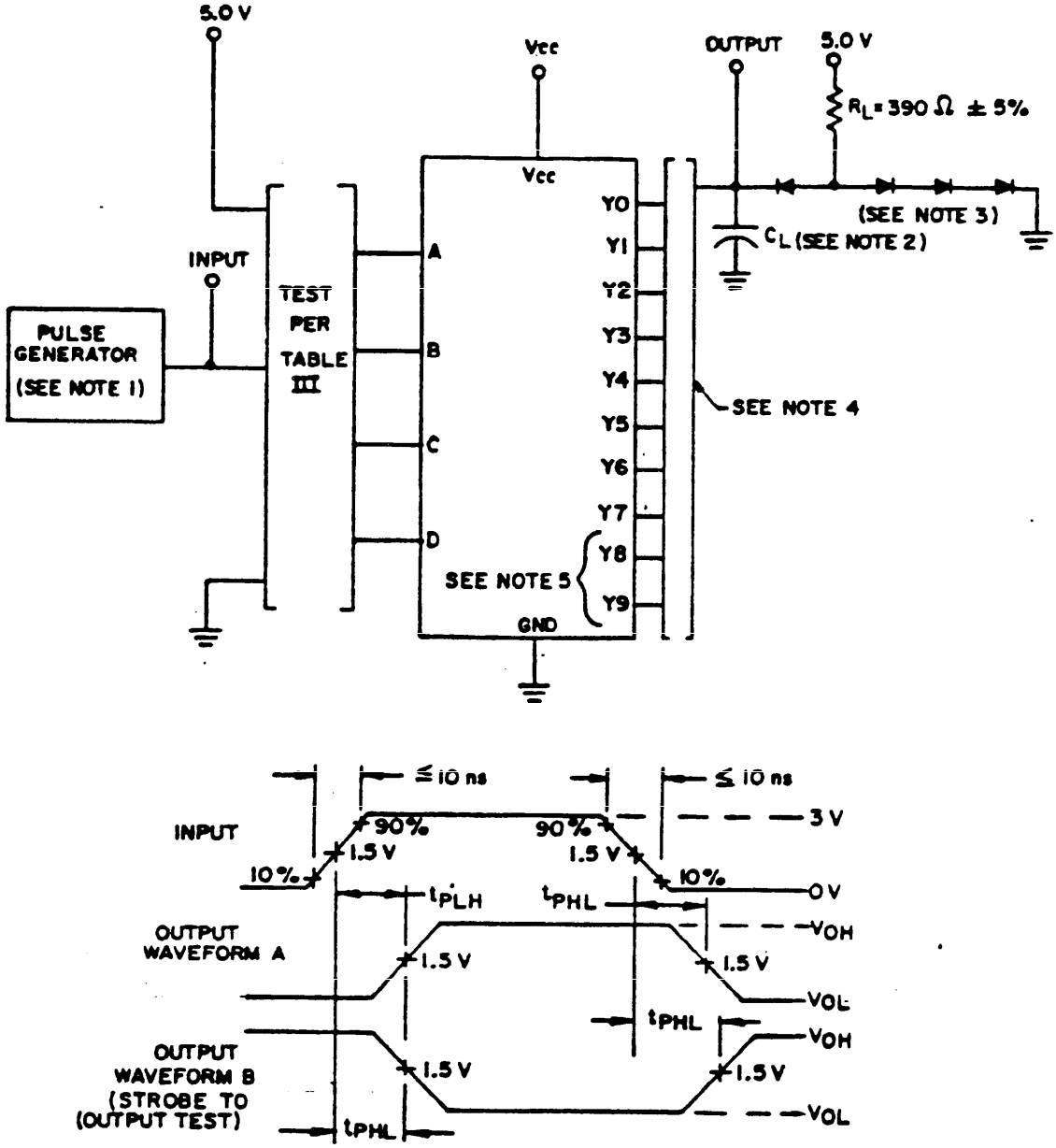
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50%, and Z_{out}=50Ω.
2. C_L = 50 pF minimum and includes probe and jig capacitance.
3. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 7. Switching test for device type 03.

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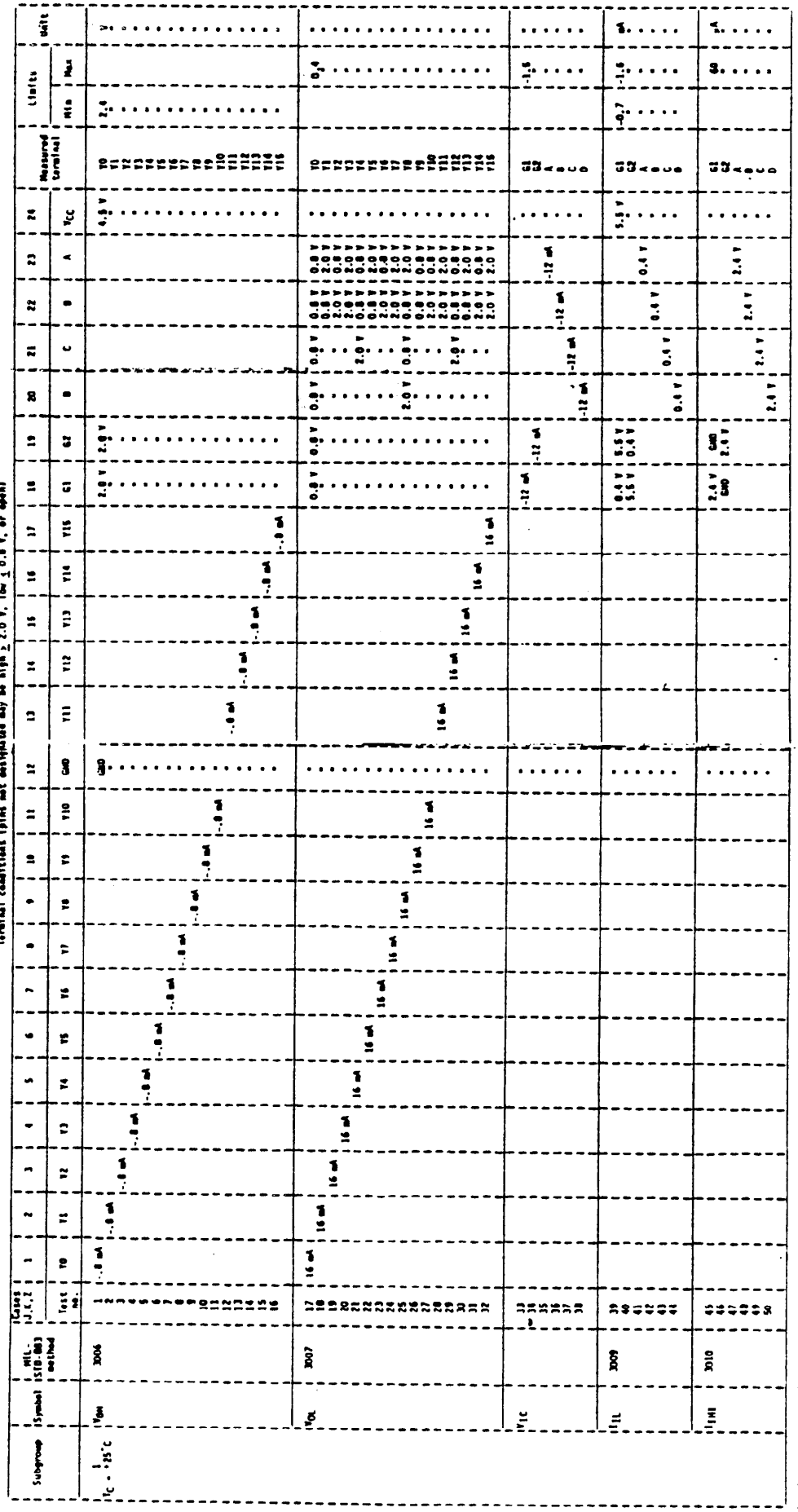
NOTES:
1. The pulse generator has the following characteristics: PRR ≤ 1 MHz, duty cycle = 50%, and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF ±10% minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.
5. Output pins Y8 and Y9 apply to device types 05 and 06 only.

FIGURE 8. Switching test for device types 04, 05, and 06.

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TABLE III. Group A inspection for device types 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)



See footnote at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits		Unit
			Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	
1 T_C = +25°C	I_{IH2}	3010	51 52 53 54 55 56												GND						5.5 V GND	GND 5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	G1 G2 A B C D		150	1/
	I_{OS}	3011	57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	5.5 V	5.5 V						Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 Y8 Y9 Y10 Y11 Y12 Y13 Y14 Y15	-20	-70	mA
	I_{CC}	3005	73																		GND	GND	GND	GND	GND	GND	GND	V_{CC}		62	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 T_C = +25°C	Truth table tests	3014	74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92	L H H H H H H H H H H H H H H H H H H H	H L H H L H H L H H L H H L H H L H H	H H L H H L H H L H H L H H L H H L H	H H H L H H L H H L H H L H H L H H L	H H H H L H H L H H L H H L H H L H H	H H H H H L H H L H H L H H L H H L H	H H H H H H L H H L H H L H H L H H L	H H H H H H H L H H L H H L H H L H H	H H H H H H H H L H H L H H L H H L H	H H H H H H H H H L H H L H H L H H L	H H H H H H H H H H L H H L H H L H H	H H H H H H H H H H H L H H L H H L H	H H H H H H H H H H H H L H H L H H L	GND	H L H H L H H L H H L H H L H H L H H	H H L H H L H H L H H L H H L H H L H	H H H L H H L H H L H H L H H L H H L	H H H H L H H L H H L H H L H H L H H	H H H H H L H H L H H L H H L H H L H	H H H H H H L H H L H H L H H L H H L	B B B B B B B B B B B B B B B B A A	B B B B B B B B B B B B B B B B B A	B B B B B B B B B B B B B B B B X X	B B A B A B A B A B A B A B A B X X	B B B A B B A B B A B B A B B A X X	B B B B A B B A B B A B B A B B A X	B B B B B A B B A B B A B B A B B A X	5.0 V	1/			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																														

See footnote at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases 1,2,C,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Unit
				Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	E1	E2	D	C	B	A	V_{CC}				
T_C = +25°C	t_{PHL1}	3003 (Fig 5)	93	OUT	OUT										GND												5.0 V	Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 Y8 Y9 Y10 Y11 Y12	5	30	ns
			94			OUT																									
			95				OUT																								
			96					OUT																							
			97						OUT																						
			98							OUT																					
			99								OUT																				
			100									OUT																			
			101										OUT																		
			102											OUT																	
			103													OUT															
			104														OUT														
			105															OUT													
	t_{PHL1}	3011 (Fig 5)	106																OUT												
			107															OUT													
T_C = +125°C	t_{PHL1}	3003 (Fig 5)	131	OUT	OUT										GND												5.0 V	Y0 Y1 Y2 Y3 Y4 Y5 Y6 Y7 Y8 Y9 Y10 Y11 Y12	5	49	ns
			132			OUT																									
			133				OUT																								
			134					OUT																							
			135						OUT																						
			136							OUT																					
			137								OUT																				
			138									OUT																			
			139										OUT																		
			140											OUT																	
			141													OUT															
			142														OUT														
			143															OUT													
			144																OUT												
			145																												
	t_{PHL2}		123	OUT																OUT											
			124																	OUT											
	t_{PLH2}		125	OUT																OUT											
			126																	OUT											
	t_{PHL3}		127							OUT	OUT	OUT																			
			128						OUT																						
			129																												
			130																												

See footnote at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be High ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases U,K,Z Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	t_{PLH1}	3003 (Fig 5)	146		OUT										GND						GND	GND	GND	GND	GND	GND	V_{CC}	Y1		83	ns
			147			OUT																						Y2			
			148				OUT																					Y3			
			149					OUT																				Y4			
			150						OUT																			Y5			
			151							OUT																		Y6			
			152								OUT																	Y7			
			153									OUT																Y8			
			154										OUT															Y9			
			155											OUT														Y10			
			156													OUT												Y11			
			157														OUT											Y12			
			158															OUT										Y13			
			159																OUT									Y14			
			160																	OUT								Y15			
	t_{PHL2}		161	Out																OUT								TO			
			162																									Y16			
	t_{PLH2}		163	OUT																OUT								TO			
			164																									Y16			
			165						OUT																			Y6			
			166							OUT																		Y7			
			167								OUT																	Y8			
			168									OUT																Y9			
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																														

See footnote at end of table.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1		2.0 V				-.8 mA	-.8 mA	GND								4.5 V	1Y0	2.4		V
			2		"				"	"	"								"	1Y1	"		"
			3		"			-.8 mA	"	"	"								"	1Y2	"		"
			4		"		-.8 mA	"	"	"	"								"	1Y3	"		"
			5				"	"	"	"	"	-.8 mA					2.0 V		"	2Y0	"		"
			6				"	"	"	"	"	"	-.8 mA				"		"	2Y1	"		"
			7				"	"	"	"	"	"	"	-.8 mA			"		"	2Y2	"		"
			8				"	"	"	"	"	"	"	"	-.8 mA		"		"	2Y3	"		"
	V_{OL}	3007	9	2.0 V	0.8 V	0.8 V				16 mA						0.8 V				1Y0		0.4	
			10	"	"	0.8 V			16 mA							2.0 V				1Y1		"	
			11	"	"	2.0 V		16 mA								0.8 V				1Y2		"	
			12			2.0 V	16 mA									2.0 V				1Y3		"	
			13			0.8 V						16 mA				0.8 V	0.8 V	0.8 V		2Y0		"	
			14			0.8 V							16 mA			2.0 V	"	"		2Y1		"	
			15			2.0 V								16 mA		0.8 V	"	"		2Y2		"	
			16			2.0 V									16 mA	2.0 V	"	"		2Y3		"	
	V_{IC}		17													-12 mA				A		-1.5	
			18			-12 mA														B		"	
			19	-12 mA																1C		"	
			20		-12 mA													-12 mA		1G		"	
			21																	2C		"	
			22														-12 mA			2G		"	
	I_{IL}	3009	23													0.4 V			5.5 V	A	-0.7	-1.6	mA
			24			0.4 V													"	B	"	"	"
			25	0.4 V															"	1C	"	"	"
			26	GND	0.4 V														"	1G	"	"	"
			27															0.4 V	"	2C	"	"	"
			28														0.4 V		"	2G	"	"	"
	I_{IH1}	3010	29													2.4 V				A		60	µA
			30			2.4 V														B		"	"
			31	2.4 V																1C		"	"
			32	5.5 V	2.4 V															1G		"	"
			33															2.4 V		2C		"	"
			34														2.4 V			2G		"	"
	I_{IH2}		35													5.5 V				A		150	
			36			5.5 V														B		"	
			37	5.5 V																1C		"	
			38	5.5 V	5.5 V															1G		"	
			39															5.5 V		2C		"	
			40														5.5 V			2G		"	
	I_{OS}	3011	41		5.5 V					GND										1Y0	-20	-55	mA
			42		"				GND											1Y1	"	"	"
			43		"			GND												1Y2	"	"	"
			44				GND													1Y3	"	"	"
			45									GND					5.5 V			2Y0	"	"	"
			46										GND				"			2Y1	"	"	"
			47											GND			"			2Y2	"	"	"
			48												GND		"			2Y3	"	"	"
	I_{CC}	3005	49		GND						"						GND	GND	"	V_{CC}		35	"

See footnote at end of table

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 1C	2 1G	3 B	4 1Y3	5 1Y2	6 1Y1	7 1Y0	8 GND	9 2Y0	10 2Y1	11 2Y2	12 2Y3	13 A	14 2G	15 2C	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
7 T_C = +25°C	Truth table test	3014	50	X	A	X	H	H	H	H	GND	H	H	H	H	X	A	X	5.0 V	1/			
			51	A	B	B	"	"	H	L	"	L	H	"	"	B	B	B	"	"			
			52	"	"	B	"	"	L	H	"	H	L	"	"	A	"	"	"	"			
			53	"	"	A	"	L	H	"	"	"	H	L	"	B	"	"	"	"			
			54	"	"	A	L	H	"	"	"	"	"	H	L	A	"	"	"	"			
			55	B	X	X	H	"	"	"	"	"	"	"	H	X	X	A	"	"			
			56	X	A	X	"	"	"	"	"	"	"	"	"	X	A	X	"	"			
			57	B	B	B	"	"	"	"	"	L	"	"	"	B	B	B	"	"			
			58	"	"	B	"	"	"	"	"	H	L	"	"	A	"	"	"	"			
			59	"	"	A	"	"	"	"	"	"	H	L	"	B	"	"	"	"			
			60	"	"	A	"	"	"	"	"	"	"	H	L	A	"	"	"	"			
			61	A	"	B	"	"	"	L	"	"	"	"	H	B	"	A	"	"			
			62	"	"	B	"	"	L	H	"	"	"	"	"	A	"	"	"	"			
			63	"	"	A	"	L	H	"	"	"	"	"	"	B	"	"	"	"			
			64	"	"	A	L	H	H	"	"	"	"	"	"	A	"	"	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																					
9 T_C = +25°C	t_{PLH1}	3003 (Fig 6)	65	5.0 V	IN	GND				OUT	GND					GND			5.0 V	1Y0	3	25	ns
			66	"	"	GND			OUT		"					5.0 V			"	1Y1	"	"	"
			67	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			68	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			69			GND					"	OUT				GND	IN	GND	"	2Y0	"	"	"
			70			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			71			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			72			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PHL1}		73	5.0 V	IN	GND				OUT	"					GND			"	1Y0	"	32	"
			74	"	"	GND			OUT		"					5.0 V			"	1Y1	"	"	"
			75	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			76	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			77			GND					"	OUT				GND	IN	GND	"	2Y0	"	"	"
			78			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			79			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			80			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PLH2}		81	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	25	"
			82			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			83	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			84			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PHL2}		85	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	32	"
			86			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			87	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			88			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PLH3}		89			GND					"	OUT				GND	GND	IN	"	2Y0	"	25	"
	t_{PHL3}		90			GND					"	OUT				GND	GND	IN	"	2Y0	"	32	"

See footnote at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1YO	GND	2YO	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
9 T_C = +25°C	t_{PLH4}	3003 (Fig 6)	91	IN	GND	GND				OUT	GND					GND			5.0 V	1YO	3	29	ns
	t_{PHL4}		92	IN	GND	GND				OUT	"					GND			"	1YO	"	35	"
	t_{PLH5}		93	5.0 V	GND	IN		OUT			"					GND			"	1Y2	"	37	"
			94			IN					"			OUT		GND	GND	GND	"	2Y2	"	"	"
			95	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			96			GND					"		OUT			IN	GND	GND	"	2Y1	"	"	"
	t_{PHL5}		97	5.0 V	GND	IN		OUT			"					GND			"	1Y2	"	"	"
			98			IN					"			OUT		GND	GND	GND	"	2Y2	"	"	"
			99	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			100			"					"		OUT			IN	GND	GND	"	2Y1	"	"	"
10 T_C = +125°C	t_{PLH1}		101	5.0 V	IN	"				OUT	"					GND			"	1YO	"	33	"
			102	"	"	"			OUT		"					5.0 V			"	1Y1	"	"	"
			103	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			104	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			105			GND					"	OUT				GND	IN	GND	"	2YO	"	"	"
			106			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			107			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			108			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PHL1}		109	5.0 V	IN	GND				OUT	"					GND			"	1YO	"	42	"
			110	"	"	GND			OUT		"					5.0 V			"	1Y1	"	"	"
			111	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			112	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			113			GND					"	OUT				GND	IN	GND	"	2YO	"	"	"
			114			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			115			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			116			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PLH2}		117	5.0 V	GND	IN				OUT	"					GND			"	1YO	"	33	"
			118			IN					"	OUT				GND	GND	GND	"	2YO	"	"	"
			119	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			120			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PHL2}		121	5.0 V	GND	IN				OUT	"					GND			"	1YO	"	42	"
			122			IN					"	OUT				GND	GND	GND	"	2YO	"	"	"
			123	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			124			5.0 V					"			OUT		IN	"	GND	"	2Y2	"	"	"
	t_{PLH3}		125			GND					"	OUT				GND	"	IN	"	2YO	"	33	"
	t_{PHL3}		126			"					"	OUT				"	"	IN	"	2YO	"	42	"
	t_{PLH4}		127	IN	GND	"				OUT	"					"			"	1YO	"	38	"
	t_{PHL4}		128	IN	GND	"				OUT	"					"			"	1YO	"	46	"

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
10 T_C = +125°C	t_{PLH5}	3003 (Fig 5)	129	5.0 V	GND	IN		OUT			GND					GND			5.0 V	1Y2	3	48	ns
			130			IN					"					GND	GND	GND	"	2Y2	"	"	"
			131	5.0 V	GND	GND			OUT		"			OUT		IN			"	1Y1	"	"	"
			132			GND					"					IN			"	2Y2	"	"	"
	t_{PHL5}		133	5.0 V	GND	IN		OUT			"					GND			"	1Y2	"	"	"
			134			IN					"			OUT		GND	GND	GND	"	2Y2	"	"	"
			135	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			136			GND					"		OUT			IN	GND	GND	"	2Y1	"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

See footnote at end of table.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1YO	GND	2YO	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
1 T_C = +25°C	I_{CEX}		1		2.0 V				5.5 V	5.5 V	GND								4.5 V	1YO		250	µA
			2		"				"	"	"								"	1Y1		"	"
			3		"			5.5 V			"								"	1Y2		"	"
			4		"		5.5 V				"								"	1Y3		"	"
			5								"	5.5 V					2.0 V		"	2YO		"	"
			6								"		5.5 V				"		"	2Y1		"	"
			7								"			5.5 V			"		"	2Y2		"	"
			8								"				5.5 V		"		"	2Y3		"	"
	V_{OL}	3007	9	2.0 V	0.8 V	0.8 V				16 mA	"					0.8 V				1YO		0.4	V
			10	"	"	0.8 V			16 mA		"					2.0 V				1Y1		"	"
			11	"	"	2.0 V		16 mA			"					0.8 V				1Y2		"	"
			12	"	"	2.0 V	16 mA				"					2.0 V				1Y3		"	"
			13			0.8 V					"	16 mA				0.8 V	0.8 V	0.8 V		2YO		"	"
			14			0.8 V					"		16 mA			2.0 V	"	"		2Y1		"	"
			15			2.0 V					"			16 mA		0.8 V	"	"		2Y2		"	"
			16			2.0 V					"					2.0 V	"	"		2Y3		"	"
	V_{IC}		17								"								"	A		-1.5	"
			18			-12 mA					"					-12 mA			"	B		"	"
			19	-12 mA							"								"	1C		"	"
			20		-12 mA						"								"	1G		"	"
			21								"							-12 mA	"	2C		"	"
			22								"						-12 mA		"	2G		"	"
	I_{IL}	3009	23								"					0.4 V			5.5 V	A	-0.7	-1.6	mA
			24			0.4 V					"								"	B	"	"	"
			25	0.4 V							"								"	1C	"	"	"
			26	GND	0.4 V						"								"	1G	"	"	"
			27								"							0.4 V	"	2C	"	"	"
			28								"						0.4 V		"	2G	"	"	"
	I_{IH1}	3010	29								"					2.4 V			"	A		60	µA
			30			2.4 V					"								"	B		"	"
			31	2.4 V							"								"	1C		"	"
			32	5.5 V	2.4 V						"								"	1G		"	"
			33								"							2.4 V	"	2C		"	"
			34								"						2.4 V		"	2G		"	"
	I_{IH2}		35								"					5.5 V			"	A		150	"
			36			5.5 V					"								"	B		"	"
			37	5.5 V							"								"	1C		"	"
			38	5.5 V	5.5 V						"								"	1G		"	"
			39								"							5.5 V	"	2C		"	"
			40								"						5.5 V		"	2G		"	"
	I_{CC}	3005	41		GND						"						GND	GND	"	V_{CC}		35	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnote at end of table.

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Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
7 T_C = +25°C	Truth table test	3014	42	X	A	X	H	H	H	H	GND	H	H	H	H	X	A	X	5.0 V	1/			
			43	A	B	B	"	H	H	L	"	L	H	"	"	B	B	B	"	"			
			44	"	"	B	"	L	L	H	"	H	L	"	"	A	"	"	"	"			
			45	"	"	A	"	L	H	"	"	"	H	"	"	B	"	"	"	"			
			46	"	"	A	L	H	"	"	"	"	"	H	L	A	"	"	"	"			
			47	B	X	X	H	"	"	"	"	"	"	"	H	X	X	A	"	"			
			48	X	A	X	"	"	"	"	"	"	"	"	"	X	A	X	"	"			
			49	B	B	B	"	"	"	"	"	L	"	"	"	B	B	B	"	"			
			50	"	"	B	"	"	"	"	"	H	L	"	"	A	"	"	"	"			
			51	"	"	A	"	"	"	"	"	"	H	L	"	B	"	"	"	"			
			52	"	"	A	"	"	"	"	"	"	"	H	L	A	"	"	"	"			
			53	A	"	B	"	"	"	L	"	"	"	"	H	B	"	A	"	"			
			54	"	"	B	"	"	L	H	"	"	"	"	"	A	"	"	"	"			
			55	"	"	A	"	L	H	"	"	"	"	"	"	B	"	"	"	"			
			56	"	"	A	L	H	H	"	"	"	"	"	"	A	"	"	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig 7)	57	5.0 V	IN	GND				OUT	GND					GND			5.0 V	1Y0	3	33	ns
			58	"	"	GND			OUT	"	"					5.0 V			"	1Y1	"	"	"
			59	"	"	5.0 V		OUT		"	"					GND			"	1Y2	"	"	"
			60	"	"	5.0 V	OUT			"	"					5.0 V			"	1Y3	"	"	"
			61			GND				"	"	OUT				GND	IN	GND	"	2Y0	"	"	"
			62			GND				"	"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			63			5.0 V				"	"			OUT		GND	"	"	"	2Y2	"	"	"
			64			5.0 V				"	"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PHL1}		65	5.0 V	IN	GND				OUT	"					GND			"	1Y0	"	40	"
			66	"	"	GND			OUT	"	"					5.0 V			"	1Y1	"	"	"
			67	"	"	5.0 V		OUT		"	"					GND			"	1Y2	"	"	"
			68	"	"	5.0 V	OUT			"	"					5.0 V			"	1Y3	"	"	"
			69			GND				"	"	OUT				GND	IN	GND	"	2Y0	"	"	"
			70			GND				"	"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			71			5.0 V				"	"			OUT		GND	"	"	"	2Y2	"	"	"
			72			5.0 V				"	"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PLH2}		73	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	33	"
			74			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			75	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			76			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PHL2}		77	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	40	"
			78			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			79	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			80			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PLH3}		81			GND					"	OUT				GND	GND	IN	"	2Y0	"	33	"
	t_{PHL3}		82			"					"	OUT				"	GND	IN	"	2Y0	"	40	"
	t_{PLH4}		83	IN	GND	"				OUT	"					"			"	1Y0	"	37	"
	t_{PHL4}		84	IN	GND	"				OUT	"					"			"	1Y0	"	43	"

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	B	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
9 T_C = +25°C	t_{PLH5}	3003 (Fig 7)	85	5.0 V	GND	IN		OUT			GND					GND			5.0 V	1Y2	3	44	ns
			86			IN					"					GND	GND	GND	"	2Y2	"	"	"
			87	5.0 V	GND	GND			OUT		"			OUT		IN			"	1Y1	"	"	"
			88			GND					"		OUT			IN	GND	GND	"	2Y1	"	"	"
	t_{PHL5}		89	5.0 V	GND	IN		OUT			"					GND			"	1Y2	"	"	"
			90			IN					"			OUT		GND	GND	GND	"	2Y2	"	"	"
			91	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			92			"					"		OUT			IN	GND	GND	"	2Y1	"	"	"
10 T_C = +125°C	t_{PLH1}		93	5.0 V	IN	"				OUT	"					GND			"	1Y0	"	43	"
			94	"	"	"			OUT		"					5.0 V			"	1Y1	"	"	"
			95	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			96	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			97			GND					"	OUT				GND	IN	GND	"	2Y0	"	"	"
			98			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			99			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			100			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PHL1}		101	5.0 V	IN	GND				OUT	"					GND			"	1Y0	"	52	"
			102	"	"	GND			OUT		"					5.0 V			"	1Y1	"	"	"
			103	"	"	5.0 V		OUT			"					GND			"	1Y2	"	"	"
			104	"	"	5.0 V	OUT				"					5.0 V			"	1Y3	"	"	"
			105			GND					"	OUT				GND	IN	GND	"	2Y0	"	"	"
			106			GND					"		OUT			5.0 V	"	"	"	2Y1	"	"	"
			107			5.0 V					"			OUT		GND	"	"	"	2Y2	"	"	"
			108			5.0 V					"				OUT	5.0 V	"	"	"	2Y3	"	"	"
	t_{PLH2}		109	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	43	"
			110			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			111	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			112			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PHL2}		113	5.0 V	GND	IN				OUT	"					GND			"	1Y0	"	52	"
			114			IN					"	OUT				GND	GND	GND	"	2Y0	"	"	"
			115	5.0 V	GND	5.0 V		OUT			"					IN			"	1Y2	"	"	"
			116			5.0 V					"			OUT		IN	GND	GND	"	2Y2	"	"	"
	t_{PLH3}		117			GND					"	OUT				GND	GND	IN	"	2Y0	"	43	"
	t_{PHL3}		118			"					"	OUT				GND	GND	IN	"	2Y0	"	52	"
	t_{PLH4}		119	IN	GND	"				OUT	"					GND			"	1Y0	"	48	"
	t_{PHL4}		120	IN	GND	"				OUT	"					GND			"	1Y0	"	56	"

See footnote at end of table.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				1C	1G	8	1Y3	1Y2	1Y1	1Y0	GND	2Y0	2Y1	2Y2	2Y3	A	2G	2C	V_{CC}				
10 T_C = +125°C	t_{PLH5}	3003 (Fig 7)	121	5.0 V	GND	IN		OUT			GND					GND			5.0 V	1Y2	3	57	ns
			122			IN					"			OUT		GND	GND		"	2Y2	"	"	"
			123	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			124			GND					"		OUT			IN	GND	GND	"	2Y1	"	"	"
	t_{PHL5}		125	5.0 V	GND	IN		OUT			"					GND			"	1Y2	"	"	"
			126			IN					"			OUT		GND	GND	GND	"	2Y2	"	"	"
			127	5.0 V	GND	GND			OUT		"					IN			"	1Y1	"	"	"
			128			GND					"		OUT			IN	GND	GND	"	2Y1	"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

See footnote at end of table.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C / Test no.	5 / 1 / A	6 / 2 / C	7 / 3 / D	8 / 4 / Y5	9 / 5 / Y6	10 / 6 / Y7	11 / 7 / GND	12 / 8 / Y4	13 / 9 / Y3	14 / 10 / Y2	1 / 11 / Y1	2 / 12 / Y0	3 / 13 / B	4 / 14 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V				GND				-.8 mA	-.8 mA	2.0 V	4.5 V	Y0	2.4		V
			2	"	"	"				"				"	"	"	"	Y1	"		"
			3	"	"	"				"			-.8 mA	"	"	"	"	Y2	"		"
			4	"	"	"				"		-.8 mA	"	"	"	"	"	Y3	"		"
			5	"	"	"				"	-.8 mA	"	"	"	"	"	"	Y4	"		"
			6	"	"	"	-.8 mA			"	"	"	"	"	"	"	"	Y5	"		"
			7	"	"	"	"	-.8 mA		"	"	"	"	"	"	"	"	Y6	"		"
			8	"	"	"	"	"	-.8 mA	"	"	"	"	"	"	"	"	Y7	"		"
	V_{OL}	3007	9	0.8 V	0.8 V	0.8 V				"				16 mA	16 mA	0.8 V	"	Y0		0.4	"
			10	2.0 V	"	"				"				"	"	0.8 V	"	Y1		"	"
			11	0.8 V	"	"				"			16 mA	"	"	2.0 V	"	Y2		"	"
			12	2.0 V	"	"				"		16 mA	"	"	"	2.0 V	"	Y3		"	"
			13	0.8 V	2.0 V	"				"	16 mA	"	"	"	"	0.8 V	"	Y4		"	"
			14	2.0 V	"	"	16 mA			"	"	"	"	"	"	0.8 V	"	Y5		"	"
			15	0.8 V	"	"	"	16 mA		"	"	"	"	"	"	2.0 V	"	Y6		"	"
			16	2.0 V	"	"	"	"	16 mA	"	"	"	"	"	"	2.0 V	"	Y7		"	"
	V_{IC}		17	-12 mA						"						-12 mA	"	A		-1.5	"
			18							"						"	"	B		"	"
			19		-12 mA					"						"	"	C		"	"
			20			-12 mA				"						"	"	D		"	"
	I_{IL}	3009	21	0.4 V						"							5.5 V	A	-.4	-1.3	mA
			22							"							"	B	"	"	"
			23		0.4 V					"						0.4 V	"	C	"	"	"
			24			0.4 V				"							"	D	"	"	"
	I_{IH1}	3010	25	2.4 V						"								A		60	µA
			26							"								B		"	"
			27		2.4 V					"						2.4 V		C		"	"
			28			2.4 V				"								D		"	"
	I_{IH2}		29	5.5 V						"								A		150	"
			30							"								B		"	"
			31		5.5 V					"						5.5 V		C		"	"
			32			5.5 V				"								D		"	"
	I_{OS}	3011	33	5.5 V	5.5 V					"								Y0	-10	-55	mA
			34	"	"					"								Y1	"	"	"
			35	"	"					"								Y2	"	"	"
			36	"	"					"								Y3	"	"	"
			37	"	"					"								Y4	"	"	"
			38	"	"		GND			"	GND	GND	GND	GND	GND	5.5 V		Y5	"	"	"
			39	"	"			GND		"								Y6	"	"	"
			40	"	"				GND	"								Y7	"	"	"
	I_{CC}	3005	41			GND				•							•			25	•

See footnote at end of table.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C Test no.	5 1 A	6 2 C	7 3 D	8 4 Y5	9 5 Y6	10 6 Y7	11 7 GND	12 8 Y4	13 9 Y3	14 10 Y2	1 11 Y1	2 12 Y0	3 13 B	4 14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Truth table test	3014	42	B	B	B	H	H	H	GND	H	H	H	H	L	B	5.0 V	1/			
			43	A	"	"	"	"	"	"	"	"	L	L	H	B	"	"			
			44	B	"	"	"	"	"	"	"	"	H	H	L	A	"	"			
			45	A	"	"	"	"	"	"	"	L	H	"	"	A	"	"			
			46	B	A	"	"	"	"	"	"	H	"	"	"	B	"	"			
			47	A	"	"	L	"	"	"	L	"	"	"	"	B	"	"			
			48	B	"	"	H	L	"	"	H	"	"	"	"	A	"	"			
			49	A	"	"	"	H	L	"	"	"	"	"	"	A	"	"			
			50	B	B	A	"	"	H	"	"	"	"	"	"	B	"	"			
			51	A	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
			52	B	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
			53	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
			54	B	A	"	"	"	"	"	"	"	"	"	"	B	"	"			
			55	A	"	"	"	"	"	"	"	"	"	"	"	B	"	"			
			56	B	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
			57	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	t_{PHL1}	3003 (Fig 8)	58	IN	GND	GND				GND					OUT	GND	5.0 V	Y0	5	37	ns
			59	5.0 V	GND	"				"				OUT		IN	"	Y1	"	"	"
			60	GND	IN	"				"			OUT			5.0 V	"	Y2	"	"	"
			61	5.0 V	GND	IN				"		OUT				5.0 V	"	Y3	"	"	"
			62	IN	5.0 V	GND				"	OUT					GND	"	Y4	"	"	"
			63	5.0 V	"	"	OUT			"						IN	"	Y5	"	"	"
			64	IN	"	"		OUT		"						5.0 V	"	Y6	"	"	"
			65	5.0 V	"	"			OUT	"						5.0 V	"	Y7	"	"	"
	t_{PLH1}		66	IN	GND	"				"					OUT	GND	"	Y0	"	"	"
			67	5.0 V	GND	"				"				OUT		IN	"	Y1	"	"	"
			68	GND	IN	"				"			OUT			5.0 V	"	Y2	"	"	"
			69	5.0 V	GND	IN				"		OUT				5.0 V	"	Y3	"	"	"
			70	IN	5.0 V	GND				"	OUT					GND	"	Y4	"	"	"
			71	5.0 V	"	"	OUT			"						IN	"	Y5	"	"	"
			72	IN	"	"		OUT		"						5.0 V	"	Y6	"	"	"
			73	5.0 V	"	IN			OUT	"						5.0 V	"	Y7	"	"	"
	t_{PHL2}		74	IN	GND	GND				"				OUT		GND	"	Y1	"	42	"
			75	5.0 V	GND	"				"		OUT				IN	"	Y3	"	"	"
			76	5.0 V	IN	"	OUT			"						GND	"	Y5	"	"	"
	t_{PLH2}		77	IN	GND	"				"				OUT		GND	"	Y1	"	"	"
			78	5.0 V	GND	"				"		OUT				IN	"	Y3	"	"	"
			79	5.0 V	IN	"	OUT			"						GND	"	Y5	"	"	"

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D	5	6	7	8	9	10	11	12	13	14	1	2	3	4	Measured terminal	Limits		Unit
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	A	C	D	Y5	Y6	Y7	GND	Y4	Y3	Y2	Y1	Y0	B	V_{CC}				
10 T_C = +125°C	t_{PHL1}	3003 (Fig 8)	80	IN	GND	GND				GND					OUT	GND	5.0 V	Y0	5	48	ns
			81	5.0 V	GND	"				"				OUT		IN	"	Y1	"	"	"
			82	GND	IN	"				"			OUT			5.0 V	"	Y2	"	"	"
			83	5.0 V	GND	IN				"		OUT				5.0 V	"	Y3	"	"	"
			84	IN	5.0 V	GND				"	OUT					GND	"	Y4	"	"	"
			85	5.0 V	"	"	OUT			"						IN	"	Y5	"	"	"
			86	IN	"	"		OUT		"						5.0 V	"	Y6	"	"	"
			87	5.0 V	"	IN			OUT	"						5.0 V	"	Y7	"	"	"
	t_{PLH1}		88	IN	GND	GND				"					OUT	GND	"	Y0	"	"	"
			89	5.0 V	GND	"				"				OUT		IN	"	Y1	"	"	"
			90	GND	IN	"				"			OUT			5.0 V	"	Y2	"	"	"
			91	5.0 V	GND	IN				"		OUT				5.0 V	"	Y3	"	"	"
			92	IN	5.0 V	GND				"	OUT					GND	"	Y4	"	"	"
			93	5.0 V	"	"	OUT			"						IN	"	Y5	"	"	"
			94	IN	"	"		OUT		"						5.0 V	"	Y6	"	"	"
			95	5.0 V	"	IN			OUT	"						5.0 V	"	Y7	"	"	"
	t_{PHL2}		96	IN	GND	GND				"				OUT		GND	"	Y1	"	53	"
			97	5.0 V	GND	"				"		OUT				IN	"	Y3	"	"	"
			98	5.0 V	IN	"	OUT			"						GND	"	Y5	"	"	"
	t_{PLH2}		99	IN	GND	"				"				OUT		GND	"	Y1	"	"	"
			100	5.0 V	GND	"				"		OUT				IN	"	Y3	"	"	"
			101	5.0 V	IN	"	OUT			"						GND	"	Y5	"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

See footnote at end of table.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be High ≥ 2.0 V, Low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				C	D	Y5	Y6	Y7	Y8	Y9	GND	Y4	Y3	Y2	Y1	Y0	B	A	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V						GND				-.8 mA	-.8 mA	2.0 V	2.0 V	4.5 V	Y0	2.4		V
			2	"	"						"			-.8 mA	"	"	"	"	"	Y1	"		"
			3	"	"						"		-.8 mA		"	"	"	"	"	Y2	"		"
			4	"	"						"	-.8 mA			"	"	"	"	"	Y3	"		"
			5	"	"						"				"	"	"	"	"	Y4	"		"
			6	"	"	-.8 mA					"				"	"	"	"	"	Y5	"		"
			7	"	"		-.8 mA				"				"	"	"	"	"	Y6	"		"
			8	"	"			-.8 mA			"				"	"	"	"	"	Y7	"		"
			9	"	"				-.8 mA		"				"	"	"	"	"	Y8	"		"
			10	"	"					-.8 mA	"				"	"	"	0.8 V	"	Y9	"		"
	V_{OL}	3007	11	0.8 V	0.8 V						"				16 mA	16 mA	0.8 V	0.8 V	"	Y0		0.4	"
			12	"	"						"						0.8 V	2.0 V	"	Y1		"	"
			13	"	"						"			16 mA			2.0 V	0.8 V	"	Y2		"	"
			14	"	"						"		16 mA				2.0 V	2.0 V	"	Y3		"	"
			15	2.0 V	"						"	16 mA					0.8 V	0.8 V	"	Y4		"	"
			16	"	"	16 mA					"						0.8 V	2.0 V	"	Y5		"	"
			17	"	"		16 mA				"						2.0 V	0.8 V	"	Y6		"	"
			18	"	"			16 mA			"						2.0 V	2.0 V	"	Y7		"	"
			19	0.8 V	2.0 V				16 mA		"						0.8 V	0.8 V	"	Y8		"	"
			20	0.8 V	2.0 V					16 mA	"						0.8 V	2.0 V	"	Y9		"	"
	V_{IC}		21								"							-12 mA	"	A		-1.5	"
			22								"						-12 mA	"	"	B		"	"
			23	-12 mA							"						"	"	"	C		"	"
			24	"	-12 mA						"						"	"	"	D		"	"
	I_{IL}	3009	25								"							0.4 V	5.5 V	A	-0.4	-1.3	mA
			26								"						0.4 V	"	"	B	"	"	"
			27	0.4 V							"						"	"	"	C	"	"	"
			28	"	0.4 V						"						"	"	"	D	"	"	"
	I_{IH1}	3010	29								"							2.4 V	"	A		60	µA
			30								"						2.4 V	"	"	B		"	"
			31	2.4 V							"						"	"	"	C		"	"
			32	"	2.4 V						"						"	"	"	D		"	"
	I_{IH2}		33								"							5.5 V	"	A		150	"
			34								"						5.5 V	"	"	B		"	"
			35	5.5 V							"						"	"	"	C		"	"
			36	"	5.5 V						"						"	"	"	D		"	"
	I_{OS}	3011	37	5.5 V	"						"				GND	GND	5.5 V	5.5 V	"	Y0	-10	-55	mA
			38	"	"						"				"	"	"	"	"	Y1	"	"	"
			39	"	"						"			GND	"	"	"	"	"	Y2	"	"	"
			40	"	"						"		GND		"	"	"	"	"	Y3	"	"	"
			41	"	"						"	GND			"	"	"	"	"	Y4	"	"	"
			42	"	"	GND					"				"	"	"	"	"	Y5	"	"	"
			43	"	"		GND				"				"	"	"	"	"	Y6	"	"	"
			44	"	"			GND			"				"	"	"	"	"	Y7	"	"	"
			45	"	"				GND		"				"	"	"	"	"	Y8	"	"	"
			46	"	"					GND	"				"	"	"	GND	"	Y9	"	"	"
	I_{CC}	3005	47		GND						"								"	V_{CC}		34	"

See footnote at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				C	D	Y5	Y6	Y7	Y8	Y9	GND	Y4	Y3	Y2	Y1	Y0	B	A	V_{CC}				
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table test	3014	48	B	B	H	H	H	H	H	GND	H	H	H	H	L	B	B	5.0 V	1/			
			49	"	"	"	"	"	"	"	"	"	"	H	L	H	B	A	"	"			
			50	"	"	"	"	"	"	"	"	"	"	L	H	L	A	B	"	"			
			51	"	"	"	"	"	"	"	"	"	L	H	H	H	A	A	"	"			
			52	A	"	"	"	"	"	"	"	L	H	"	"	"	B	B	"	"			
			53	"	"	L	"	"	"	"	"	H	"	"	"	"	B	A	"	"			
			54	"	"	H	L	"	"	"	"	"	"	"	"	"	A	B	"	"			
			55	"	"	"	H	L	"	"	"	"	"	"	"	"	A	A	"	"			
			56	B	A	"	"	H	L	"	"	"	"	"	"	"	B	B	"	"			
			57	"	"	"	"	"	H	L	"	"	"	"	"	"	B	A	"	"			
			58	"	"	"	"	"	L	H	"	"	"	"	"	"	A	B	"	"			
			59	"	"	"	"	"	H	L	"	"	"	"	"	"	A	A	"	"			
			60	A	"	"	"	"	L	H	"	"	"	"	"	"	B	B	"	"			
			61	"	"	"	"	"	H	L	"	"	"	"	"	"	B	A	"	"			
			62	"	"	"	"	"	L	H	"	"	"	"	"	"	A	B	"	"			
			63	"	"	"	"	"	H	L	"	"	"	"	"	"	A	A	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PHL1}	3003 (Fig 8)	64	GND	GND						GND					OUT	GND	IN	5.0 V	Y0	5	37	ns
			65	GND	"						"				OUT		IN	5.0 V	"	Y1	"	"	
			66	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	
			67	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	
			68	5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	
			69	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	
			70	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	
			71	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	
			72	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	
			73	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	
	t_{PLH1}		74	"	GND						"					OUT	GND	IN	"	Y0	"	"	
			75	"	"						"				OUT		IN	5.0 V	"	Y1	"	"	
			76	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	
			77	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	
			78	-5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	
			79	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	
			80	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	
			81	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	
			82	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	
			83	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	
	t_{PHL2}		84	"	GND						"						GND	IN	"	Y1	"	42	
			85	"	"						"				OUT		IN	5.0 V	"	Y3	"	"	
			86	IN	"	OUT					"						GND	"	"	Y5	"	"	
			87	5.0 V	IN					OUT	"						5.0 V	"	"	Y9	"	"	
	t_{PLH2}		88	GND	GND						"						GND	IN	"	Y1	"	"	
			89	GND	"						"				OUT		IN	5.0 V	"	Y3	"	"	
			90	IN	"	OUT					"						GND	"	"	Y5	"	"	
			91	5.0 V	IN					OUT	"						5.0 V	"	"	Y9	"	"	

See footnote at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1 C	2 D	3 Y5	4 Y6	5 Y7	6 Y8	7 Y9	8 GND	9 Y4	10 Y3	11 Y2	12 Y1	13 Y0	14 B	15 A	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	t_{PHL1}	3003 (Fig 8)	92	GND	GND						GND					OUT	GND	IN	5.0 V	Y0	5	48	ns
			93	GND	"						"				OUT		IN	5.0 V	"	Y1	"	"	"
			94	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	"
			95	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	"
			96	5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	"
			97	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	"
			98	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	"
			99	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	"
			100	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	"
			101	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	"
	t_{PLH1}		102	"	GND						"					OUT	GND	IN	"	Y0	"	"	"
			103	"	"						"				OUT		IN	5.0 V	"	Y1	"	"	"
			104	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	"
			105	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	"
			106	5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	"
			107	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	"
			108	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	"
			109	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	"
			110	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	"
			111	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	"
	t_{PHL2}		112	"	GND						"						GND	IN	"	Y1	"	53	"
			113	"	"						"						IN	5.0 V	"	Y3	"	"	"
			114	IN	"	OUT					"						GND	"	"	Y5	"	"	"
			115	5.0 V	IN					OUT	"						5.0 V	"	"	Y9	"	"	"
	t_{PLH2}		116	GND	GND						"						GND	IN	"	Y1	"	"	"
			117	GND	"						"						IN	5.0 V	"	Y3	"	"	"
			118	IN	"	OUT					"						GND	"	"	Y5	"	"	"
			119	5.0 V	IN					OUT	"						5.0 V	"	"	Y9	"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

See footnote at end of table.

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				C	D	Y5	Y6	Y7	Y8	Y9	GND	Y4	Y3	Y2	Y1	Y0	B	A	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V						GND					-.8 mA	2.0 V	2.0 V	4.5 V	Y0	2.4		V
			2	"	"						"				-.8 mA	"	"	"	"	Y1	"		"
			3	"	"						"			-.8 mA	"	"	"	"	"	Y2	"		"
			4	"	"						"		-.8 mA	"	"	"	"	"	"	Y3	"		"
			5	"	"						"	-.8 mA	"	"	"	"	"	"	"	Y4	"		"
			6	"	"	-.8 mA					"	"	"	"	"	"	"	"	"	Y5	"		"
			7	"	"		-.8 mA				"	"	"	"	"	"	"	"	"	Y6	"		"
			8	"	"			-.8 mA			"	"	"	"	"	"	"	"	"	Y7	"		"
			9	"	"				-.8 mA		"	"	"	"	"	"	"	"	"	Y8	"		"
			10	"	"					-.8 mA	"	"	"	"	"	"	"	"	"	Y9	"		"
	V_{OL}	3007	11	0.8 V	0.8 V						"				16 mA	16 mA	0.8 V	0.8 V	"	Y0		0.4	"
			12	"	"						"						0.8 V	2.0 V	"	Y1		"	"
			13	"	"						"			16 mA			2.0 V	0.8 V	"	Y2		"	"
			14	"	"						"		16 mA				2.0 V	2.0 V	"	Y3		"	"
			15	2.0 V	"						"	16 mA					0.8 V	0.8 V	"	Y4		"	"
			16	"	"	16 mA					"						0.8 V	2.0 V	"	Y5		"	"
			17	"	"		16 mA				"						2.0 V	0.8 V	"	Y6		"	"
			18	"	"			16 mA			"						2.0 V	2.0 V	"	Y7		"	"
			19	0.8 V	2.0 V				16 mA		"						0.8 V	0.8 V	"	Y8		"	"
			20	0.8 V	2.0 V					16 mA	"						0.8 V	2.0 V	"	Y9		"	"
	V_{IC}		21								"							-12 mA	"	A		-1.5	"
			22								"						-12 mA		"	B		"	"
			23	-12 mA							"								"	C		"	"
			24		-12 mA						"								"	D		"	"
	I_{IL}	3009	25								"							0.4 V	5.5 V	A	-0.7	-1.6	mA
			26								"						0.4 V		"	B	"	"	"
			27	0.4 V							"								"	C	"	"	"
			28		0.4 V						"								"	D	"	"	"
	I_{IH1}	3010	29								"							2.4 V	"	A		60	µA
			30								"						2.4 V		"	B		"	"
			31	2.4 V							"								"	C		"	"
			32		2.4 V						"								"	D		"	"
	I_{IH2}		33								"							5.5 V	"	A		150	"
			34								"						5.5 V		"	B		"	"
			35	5.5 V							"								"	C		"	"
			36		5.5 V						"								"	D		"	"
	I_{OS}	3011	37	5.5 V	"						"				GND	GND	5.5 V	5.5 V	"	Y0	-10	-55	mA
			38	"	"						"			GND			"	"	"	Y1	"	"	"
			39	"	"						"		GND				"	"	"	Y2	"	"	"
			40	"	"						"	GND					"	"	"	Y3	"	"	"
			41	"	"						"						"	"	"	Y4	"	"	"
			42	"	"	GND					"						"	"	"	Y5	"	"	"
			43	"	"		GND				"						"	"	"	Y6	"	"	"
			44	"	"			GND			"						"	"	"	Y7	"	"	"
			45	"	"				GND		"						"	"	"	Y8	"	"	"
			46	"	"					GND	"						"	"	"	Y9	"	"	"
	I_{CC}	3005	47		GND						"								"	V_{CC}		44	"

See footnote at end of table.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				C	D	Y5	Y6	Y7	Y8	Y9	GND	Y4	Y3	Y2	Y1	Y0	B	A	V_{CC}				
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table test	3014	48	B	B	H	H	H	H	H	GND	H	H	H	H	L	B	B	5.0 V	1/			
			49	"	"	"	"	"	"	"	"	"	"	H	L	H	B	A	"	"			
			50	"	"	"	"	"	"	"	"	"	"	L	H	"	A	B	"	"			
			51	"	"	"	"	"	"	"	"	"	L	H	"	"	A	A	"	"			
			52	A	"	"	"	"	"	"	"	L	H	"	"	"	B	B	"	"			
			53	"	"	L	"	"	"	"	"	H	"	"	"	"	B	A	"	"			
			54	"	"	H	L	"	"	"	"	"	"	"	"	"	A	B	"	"			
			55	"	"	"	H	L	"	"	"	"	"	"	"	"	A	A	"	"			
			56	B	A	"	"	H	L	"	"	"	"	"	"	"	B	B	"	"			
			57	"	"	"	"	"	H	L	"	"	"	"	"	"	B	A	"	"			
			58	"	"	"	"	"	"	H	"	"	"	"	"	"	A	B	"	"			
			59	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
			60	A	"	"	"	"	"	"	"	"	"	"	"	"	B	B	"	"			
			61	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"	"			
			62	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"	"			
			63	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PHL1}	3003 (Fig 8)	64	GND	GND						GND					OUT	GND	IN	5.0 V	Y0	5	37	ns
			65	GND	"						"						IN	5.0 V	"	Y1	"	"	
			66	IN	"						"			OUT	OUT		5.0 V	GND	"	Y2	"	"	
			67	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	
			68	5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	
			69	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	
			70	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	
			71	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	
			72	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	
			73	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	
	t_{PLH1}		74	"	GND						"					OUT	GND	IN	"	Y0	"	"	
			75	"	"						"				OUT		IN	5.0 V	"	Y1	"	"	
			76	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	
			77	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	
			78	5.0 V	GND						"	OUT					GND	IN	"	Y4	"	"	
			79	"	GND	OUT					"						IN	5.0 V	"	Y5	"	"	
			80	"	IN		OUT				"						5.0 V	GND	"	Y6	"	"	
			81	"	IN			OUT			"						5.0 V	5.0 V	"	Y7	"	"	
			82	GND	5.0 V				OUT		"						GND	IN	"	Y8	"	"	
			83	"	5.0 V					OUT	"						IN	5.0 V	"	Y9	"	"	
	t_{PHL2}		84	"	GND						"					OUT	GND	IN	"	Y1	"	42	
			85	"	"						"				OUT		IN	5.0 V	"	Y3	"	"	
			86	IN	"	OUT					"			OUT			GND	5.0 V	"	Y5	"	"	
			87	GND	IN					OUT	"						5.0 V	GND	"	Y9	"	"	
	t_{PLH2}		88	"	GND						"					OUT	GND	IN	"	Y1	"	"	
			89	"	"						"				OUT		IN	5.0 V	"	Y3	"	"	
			90	IN	"	OUT					"			OUT			GND	5.0 V	"	Y5	"	"	
			91	GND	IN					OUT	"						5.0 V	GND	"	Y9	"	"	

See footnote at end of table.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				C	D	Y5	Y6	Y7	Y8	Y9	GND	Y4	Y3	Y2	Y1	YO	B	A	V_{CC}				
10 T_C = +125°C	t_{PHL1}	3003 (Fig 8)	92	GND	GND						GND					OUT	GND	IN	5.0 V	YO	5	48	ns
			93	GND	"						"				OUT		IN	5.0 V	"	Y1	"	"	"
			94	IN	"						"			OUT			5.0 V	GND	"	Y2	"	"	"
			95	GND	IN						"		OUT				5.0 V	5.0 V	"	Y3	"	"	"
			96	5.0 V	GND	OUT					"	OUT					GND	IN	"	Y4	"	"	"
			97	"	GND		OUT				"						IN	5.0 V	"	Y5	"	"	"
			98	"	IN			OUT			"						5.0 V	GND	"	Y6	"	"	"
			99	"	IN				OUT		"						5.0 V	5.0 V	"	Y7	"	"	"
			100	GND	5.0 V					OUT	"						GND	IN	"	Y8	"	"	"
			101	"	5.0 V						"						IN	5.0 V	"	Y9	"	"	"
	t_{PLH1}		102	"	GND												GND	IN	"	YO	"	"	"
			103	"	"												IN	5.0 V	"	Y1	"	"	"
			104	IN	"												5.0 V	GND	"	Y2	"	"	"
			105	GND	IN												5.0 V	5.0 V	"	Y3	"	"	"
			106	5.0 V	GND	OUT						OUT					GND	IN	"	Y4	"	"	"
			107	"	GND		OUT						OUT				IN	5.0 V	"	Y5	"	"	"
			108	"	IN			OUT						OUT			5.0 V	GND	"	Y6	"	"	"
			109	"	IN				OUT						OUT		5.0 V	5.0 V	"	Y7	"	"	"
			110	GND	5.0 V					OUT							GND	IN	"	Y8	"	"	"
			111	"	5.0 V												IN	5.0 V	"	Y9	"	"	"
	t_{PHL2}		112	"	GND												GND	IN	"	Y1	"	53	"
			113	"	"												IN	5.0 V	"	Y3	"	"	"
			114	IN	"	OUT							OUT		OUT		GND	"	"	Y5	"	"	"
			115	GND	IN					OUT							"	"	"	Y9	"	"	"
	t_{PLH2}		116	"	GND												"	IN	"	Y1	"	"	"
			117	"	"												IN	5.0 V	"	Y3	"	"	"
			118	IN	"	OUT							OUT		OUT		GND	"	"	Y5	"	"	"
			119	GND	IN					OUT							GND	"	"	Y9	"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ Inputs: A = 2 V minimum and B = 0.8 V maximum.
Input: X = A or B
Outputs: Output voltages shall be either:
(a) H = 2.4 V and L = 0.4 V when using a high speed checker double comparator.
(b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.

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3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line, not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C, minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.5) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table I herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
V_{IN}	Voltage level at an input terminal
V_{IC}	Input clamp voltage
I_{IN}	Current flowing into an input terminal

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6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54154, 9311
02	54155
03	54156
04	8250
05	8251
06	8252, 9301

6.7 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/152B	MICROCIRCUITS, DIGITAL, TTL, DATA DECODERS/DE-MULTIPLEXERS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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